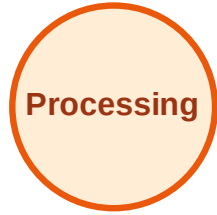


BFS Traversal

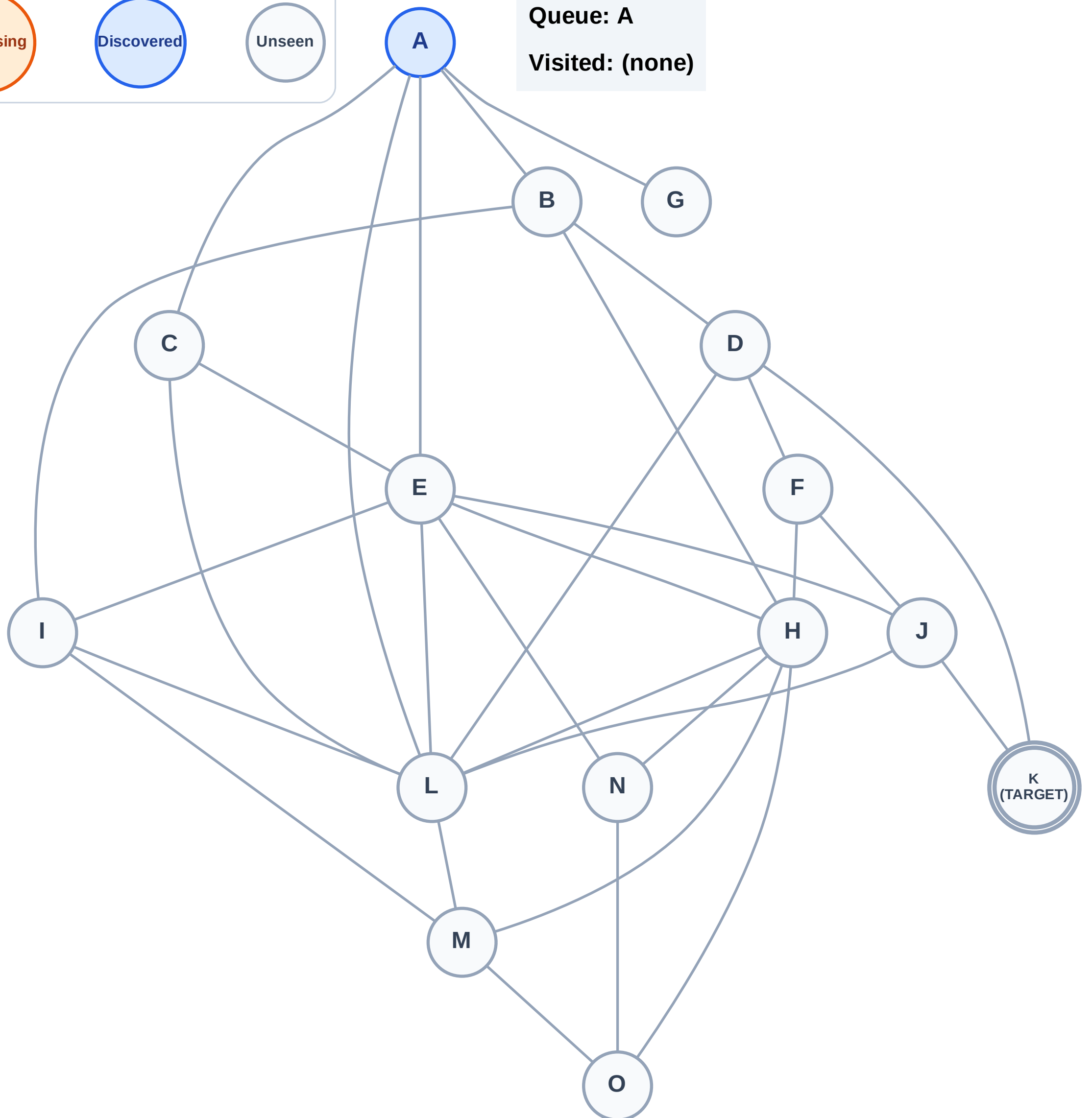
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

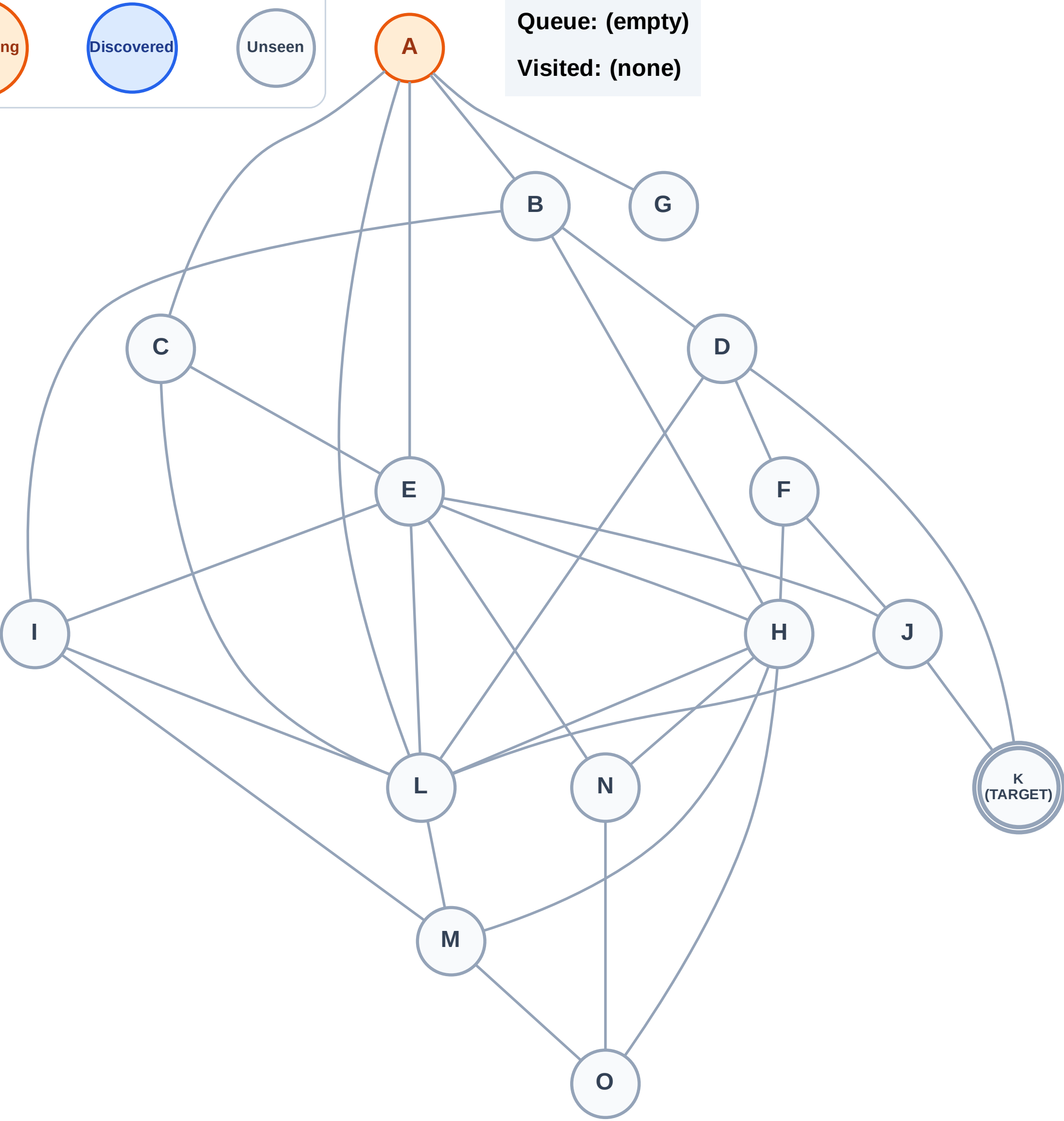
Complete

Processing

Discovered

Unseen

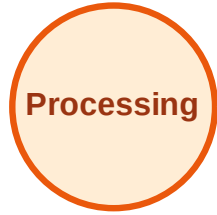
Queue: (empty)
Visited: (none)



BFS Traversal

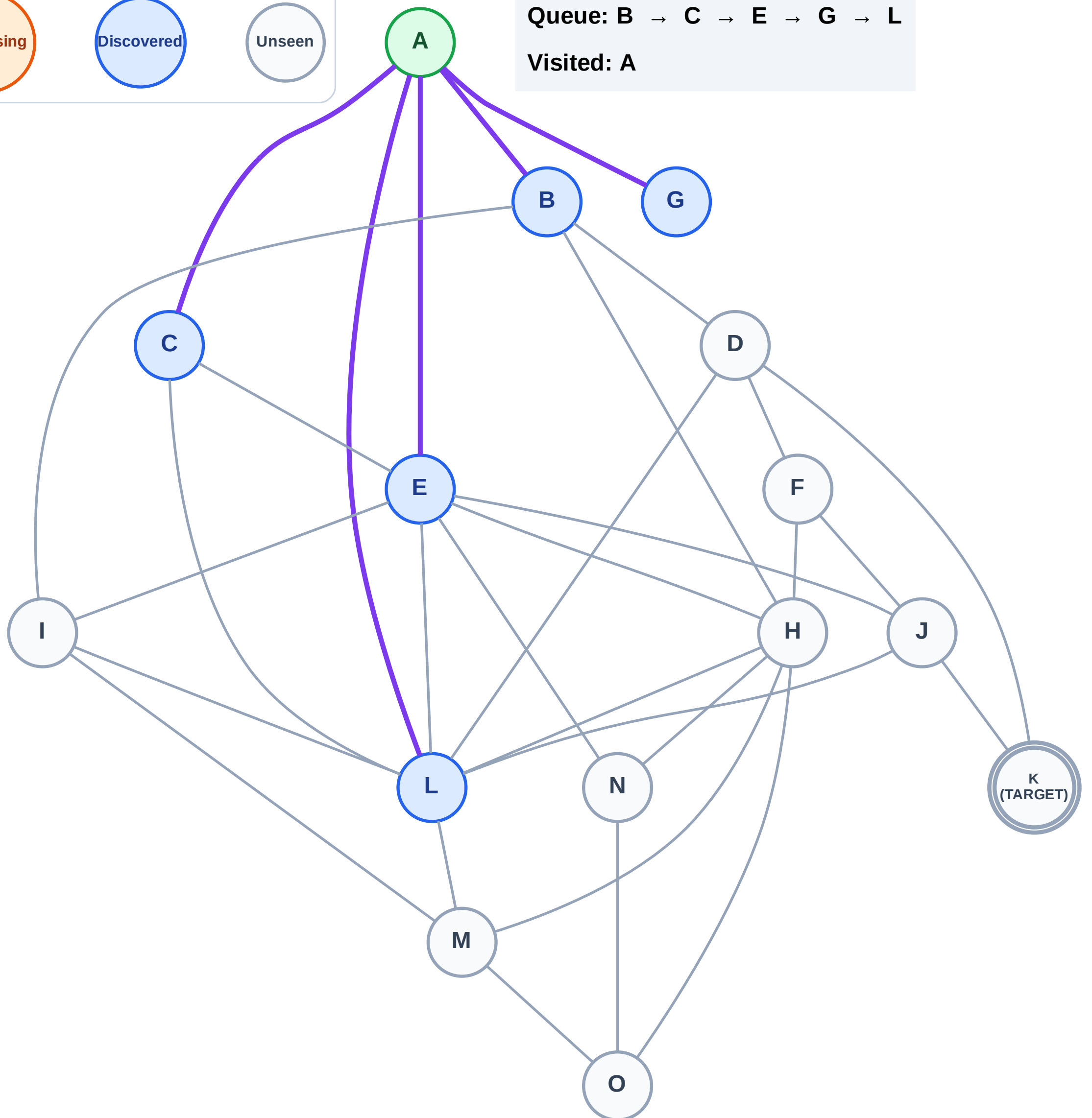
Step 3: Discover B, C, E, G, L from A

Legend



Queue: B → C → E → G → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

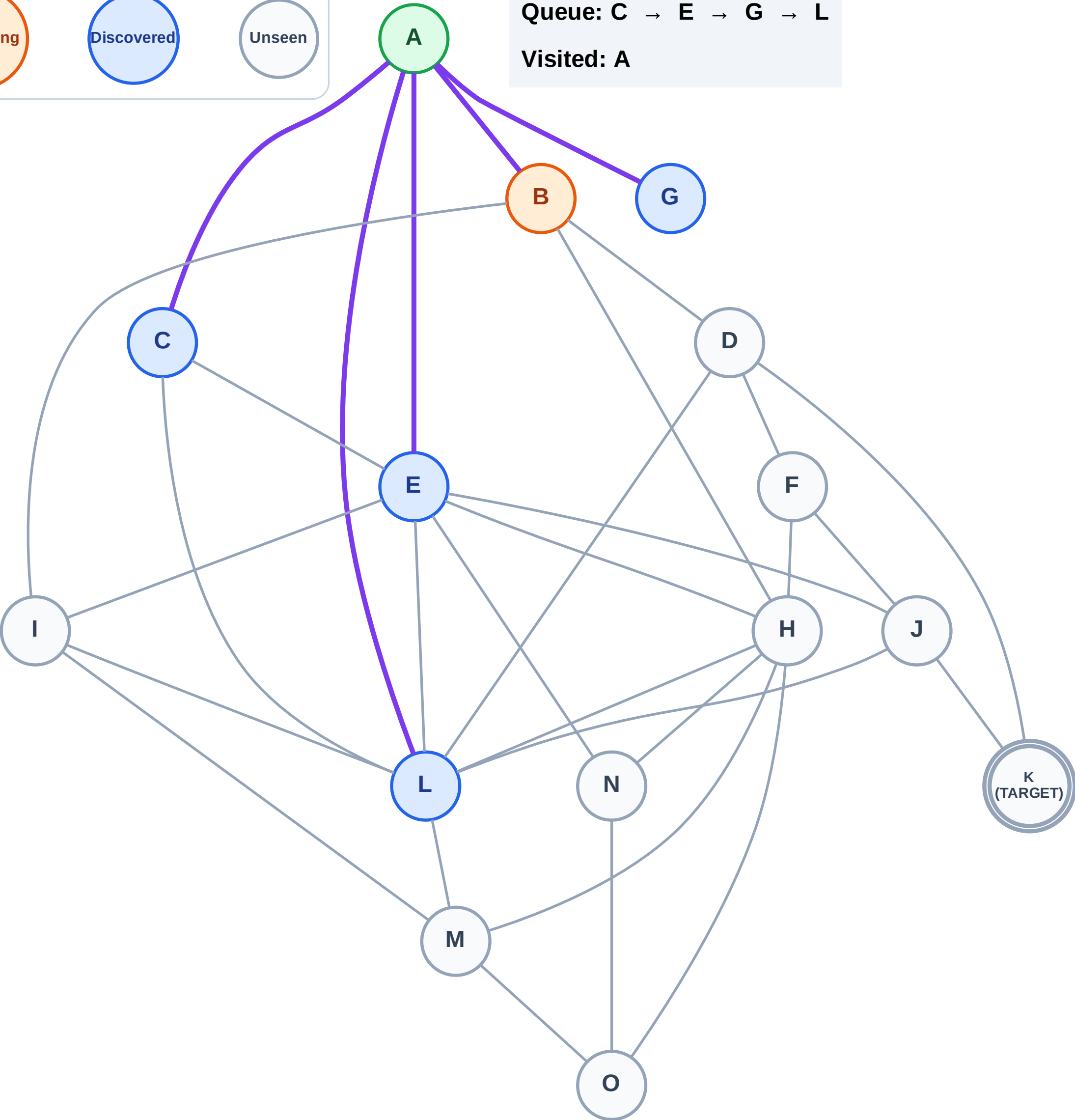
Processing

Discovered

Unseen

Queue: C → E → G → L

Visited: A



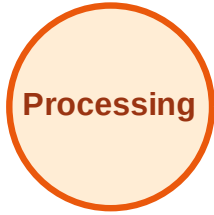
BFS Traversal

Step 5: Discover D, H, I from B

Legend



Complete



Processing



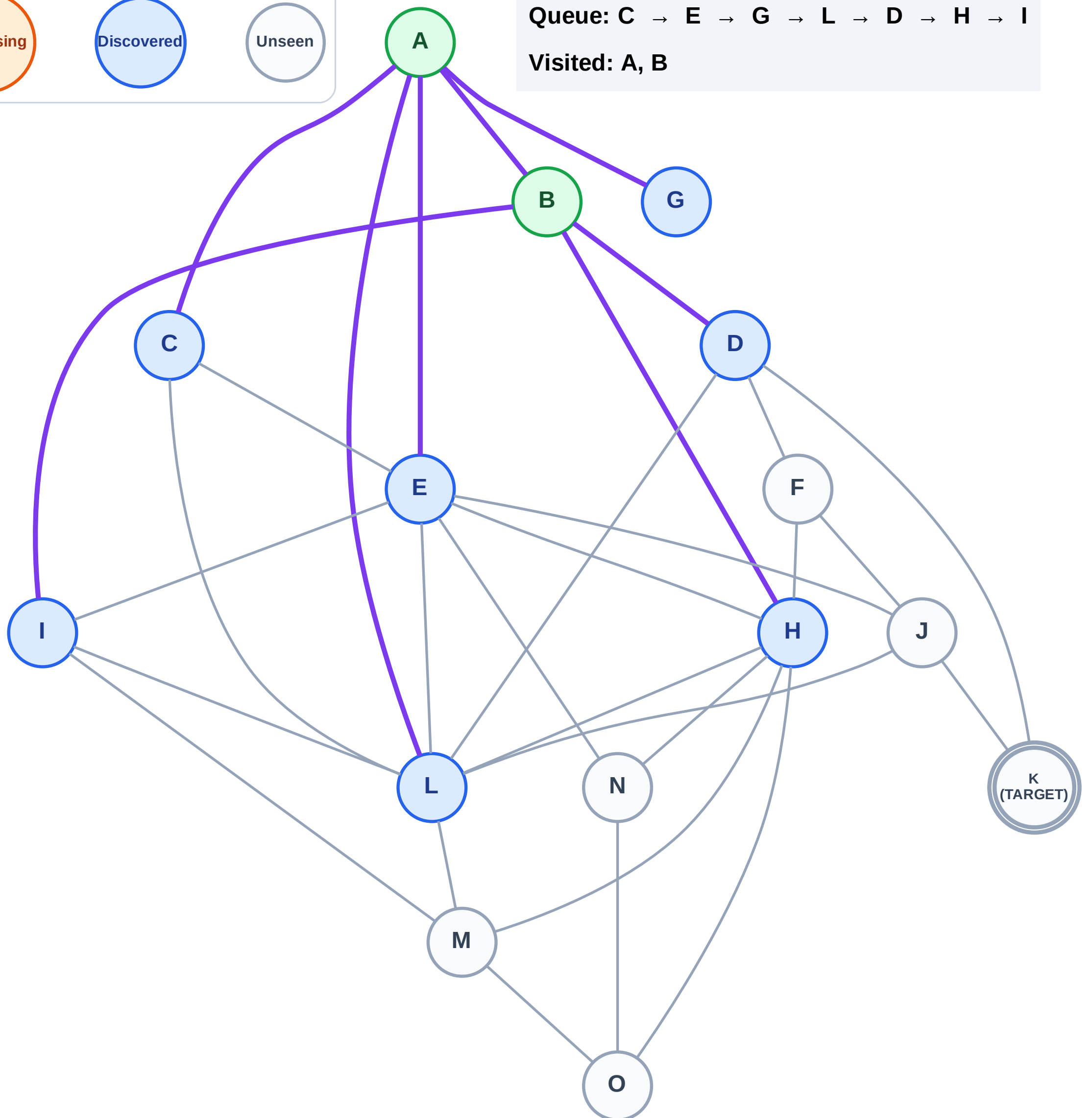
Discovered



Unseen

Queue: C → E → G → L → D → H → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

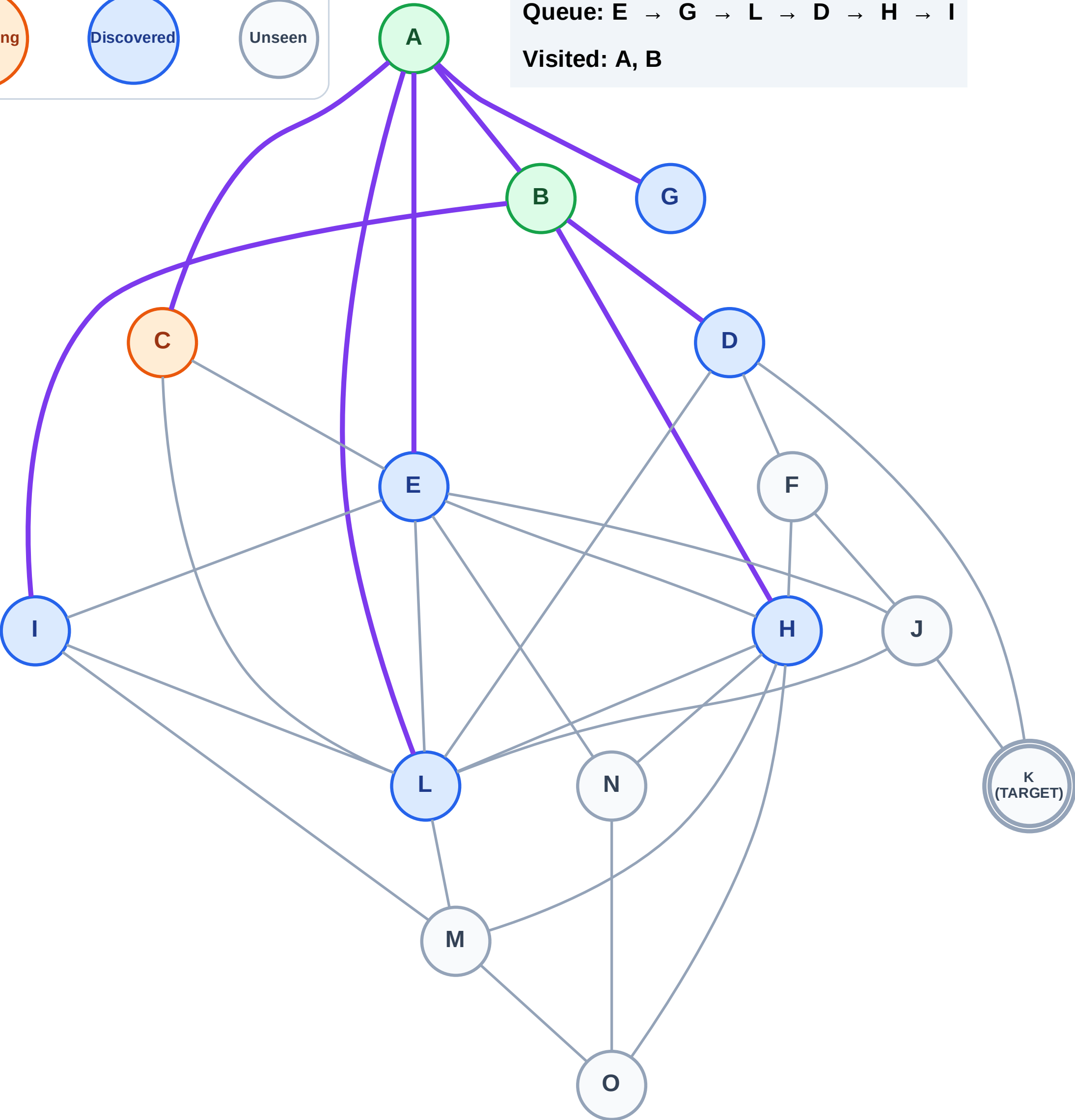
Processing

Discovered

Unseen

Queue: E → G → L → D → H → I

Visited: A, B



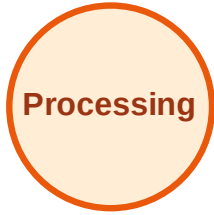
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



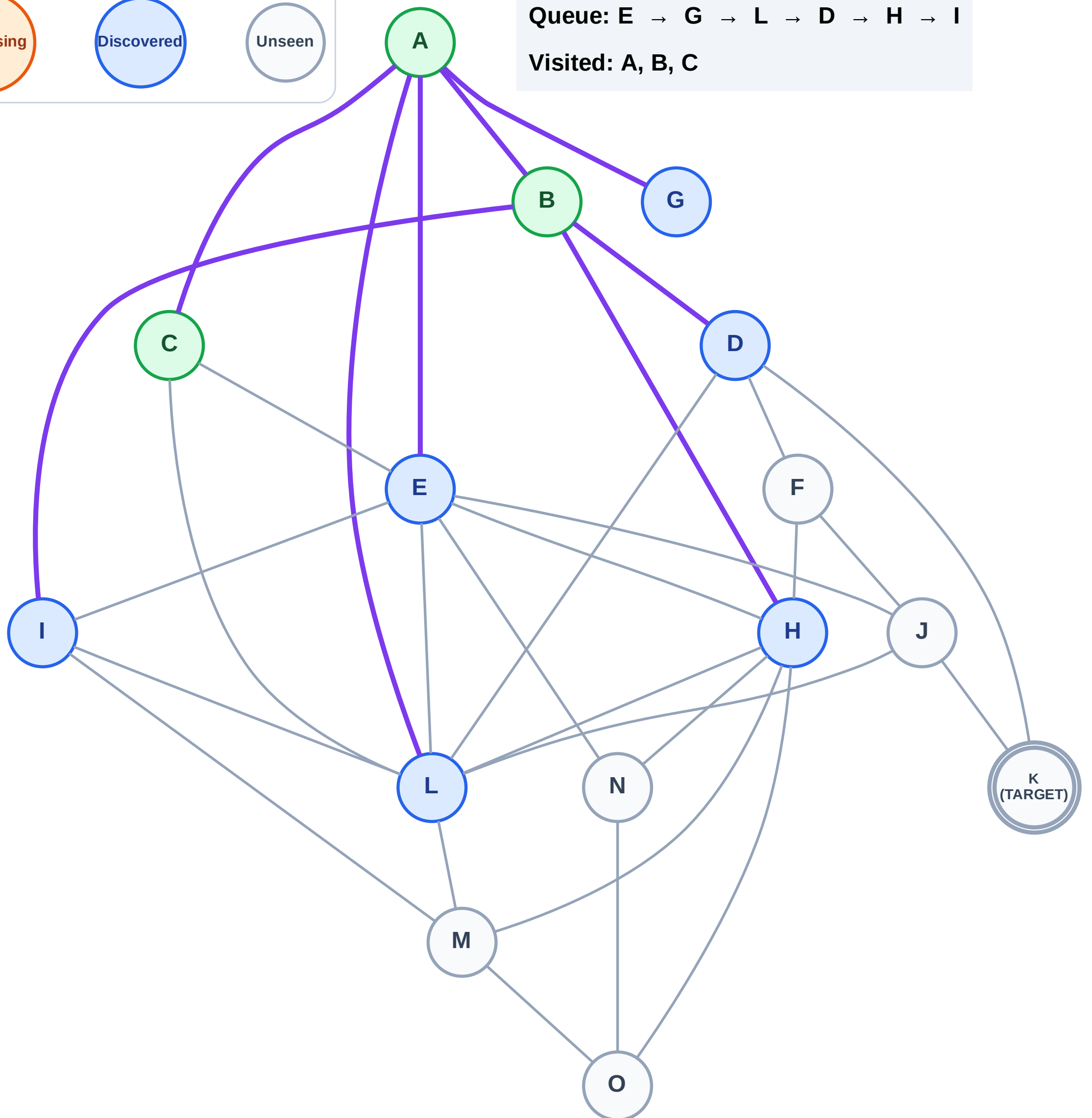
Discovered



Unseen

Queue: E → G → L → D → H → I

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

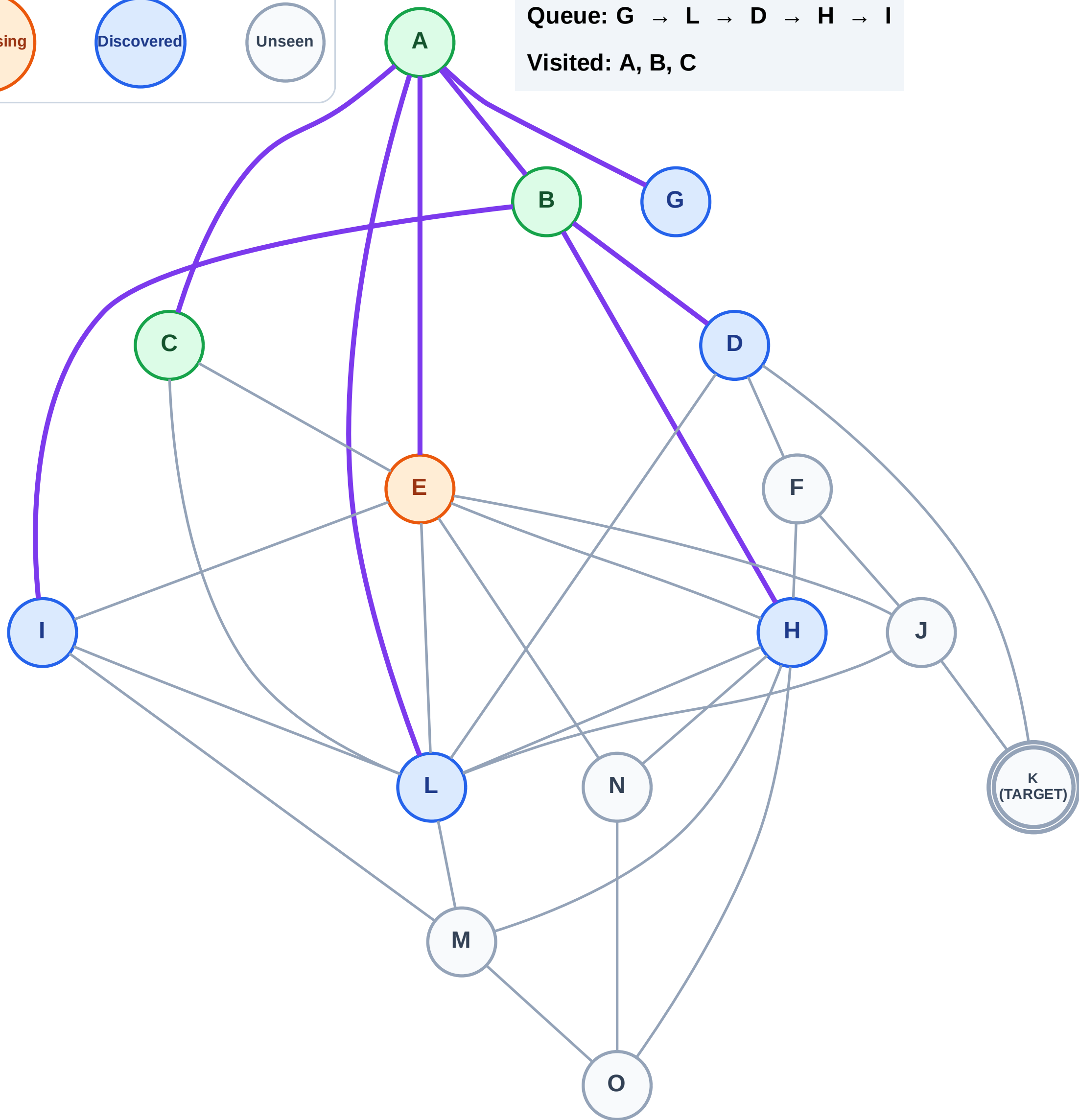
Processing

Discovered

Unseen

Queue: G → L → D → H → I

Visited: A, B, C



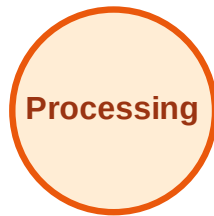
BFS Traversal

Step 9: Discover J, N from E

Legend



Complete



Processing



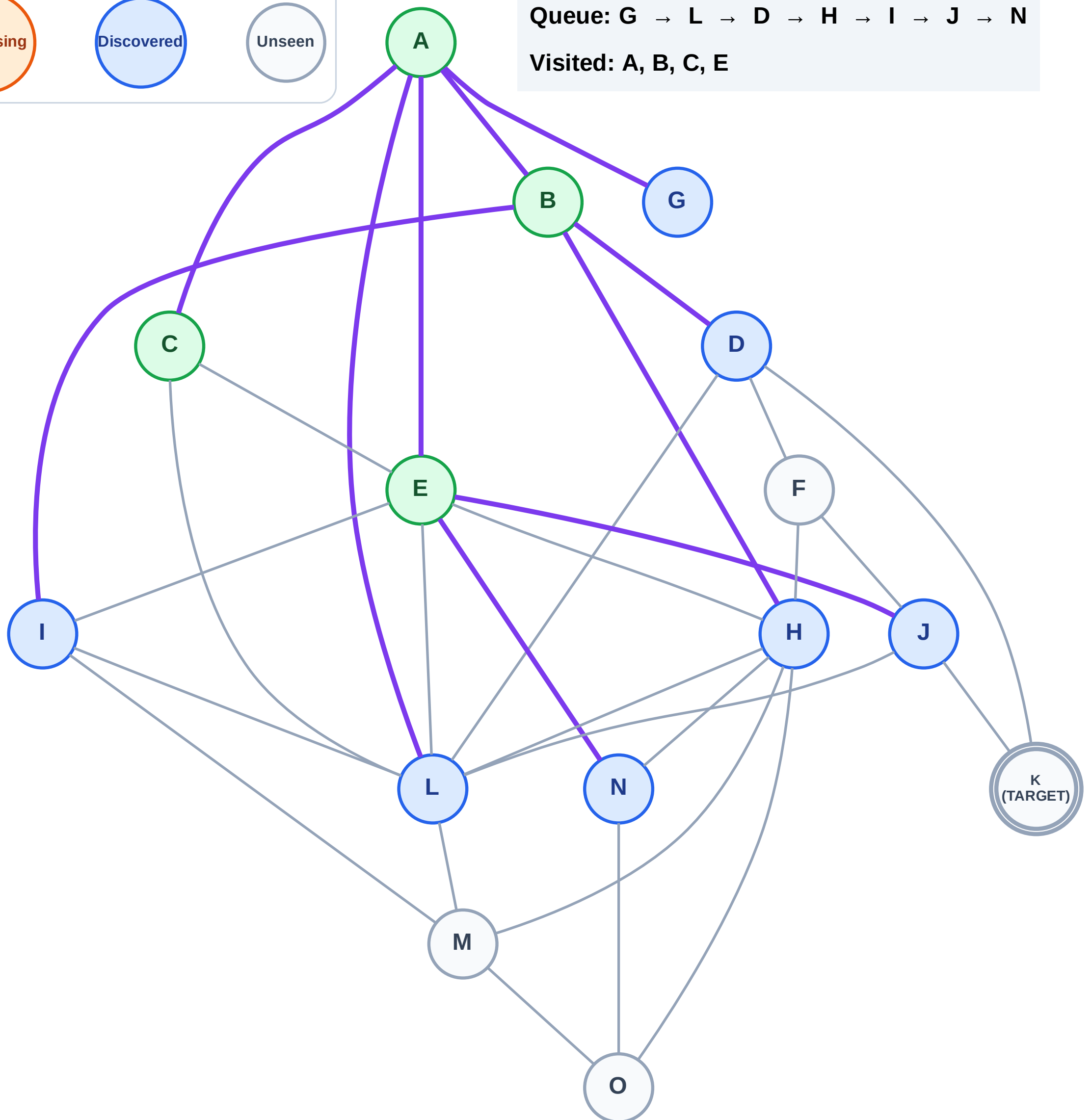
Discovered



Unseen

Queue: G → L → D → H → I → J → N

Visited: A, B, C, E



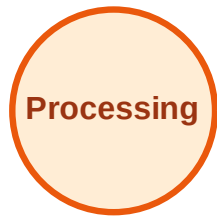
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



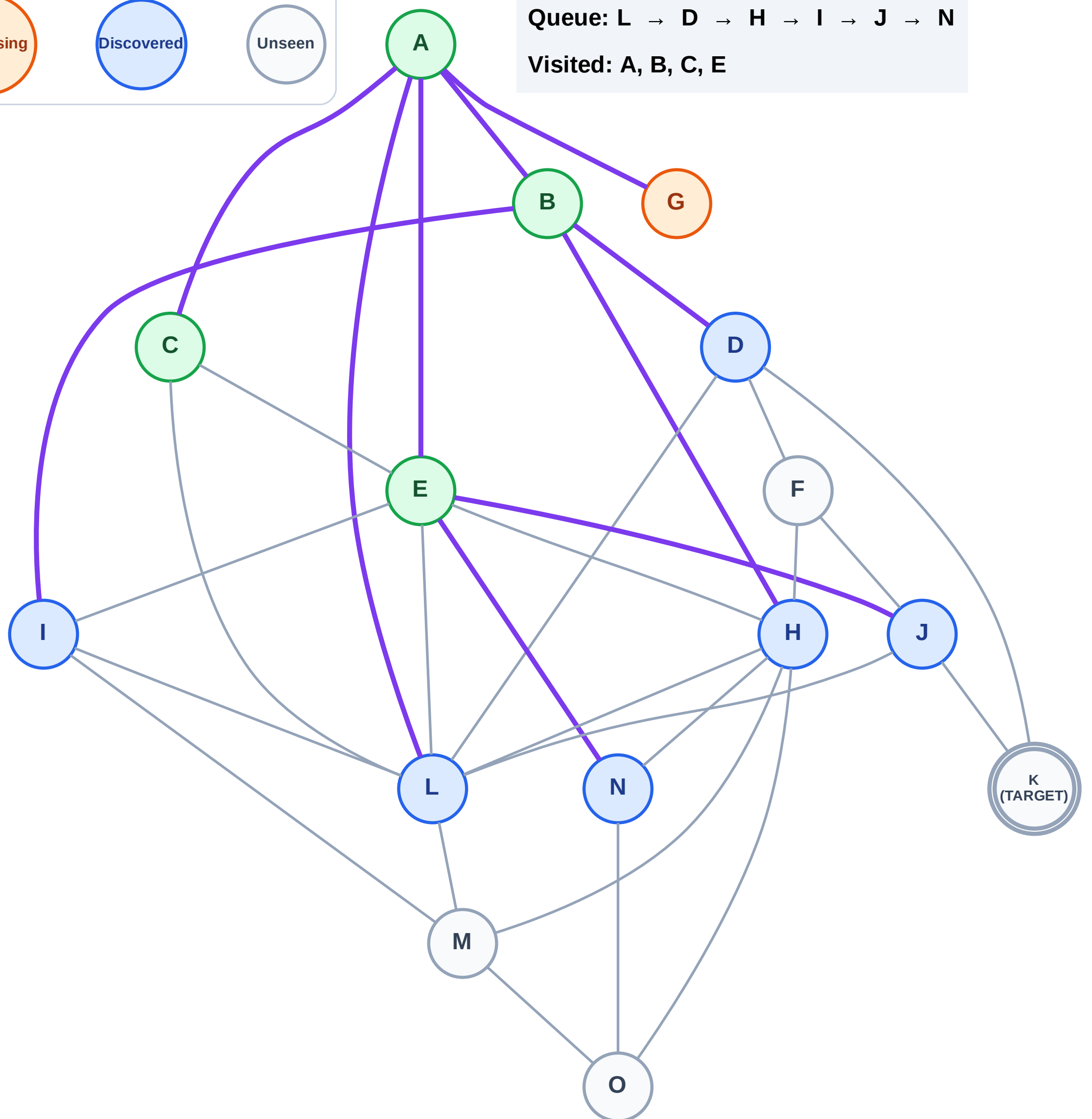
Discovered



Unseen

Queue: L → D → H → I → J → N

Visited: A, B, C, E



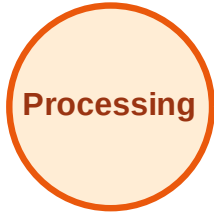
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



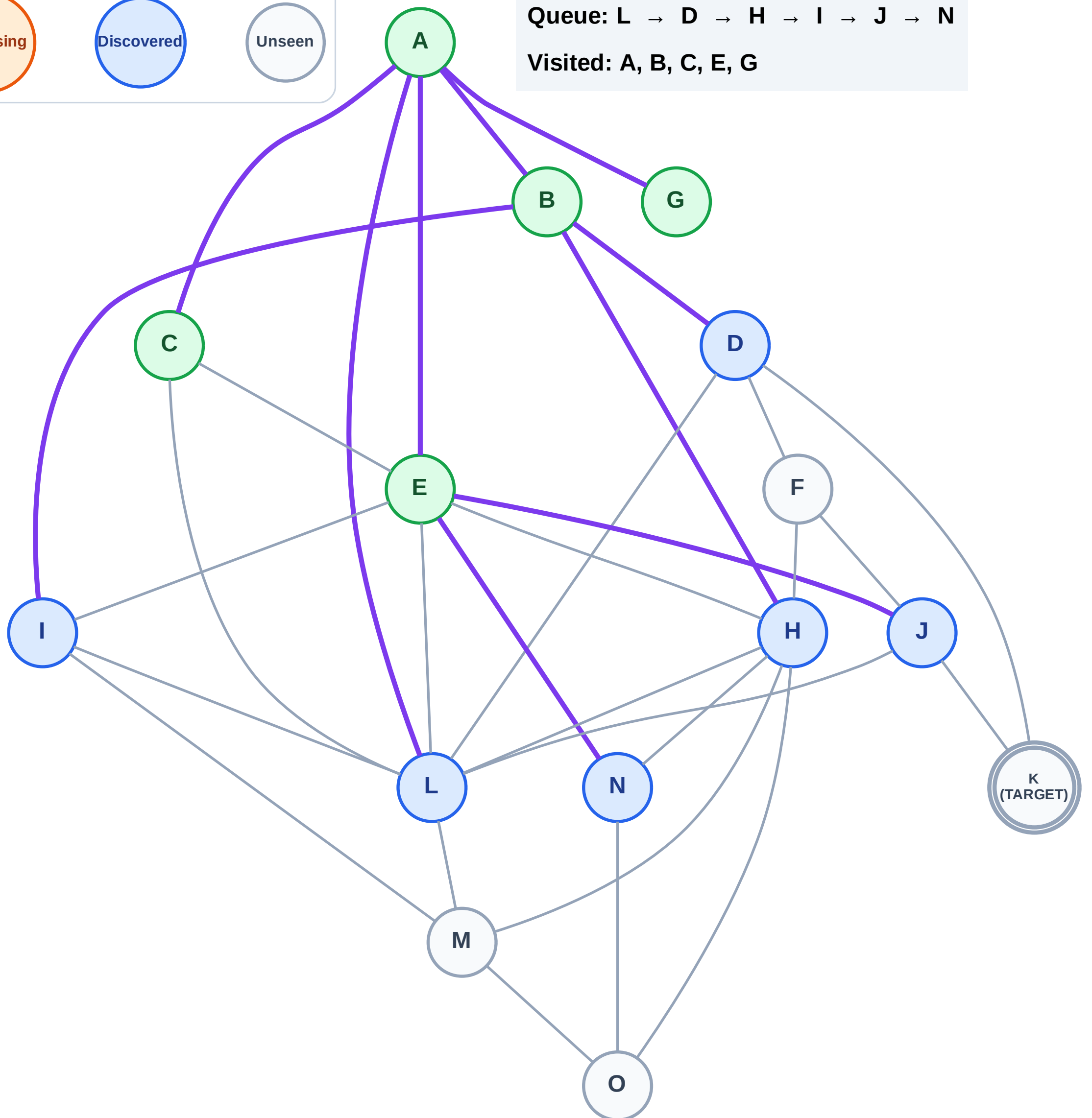
Discovered



Unseen

Queue: L → D → H → I → J → N

Visited: A, B, C, E, G



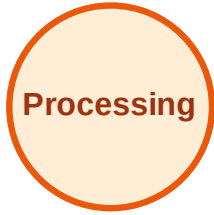
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing



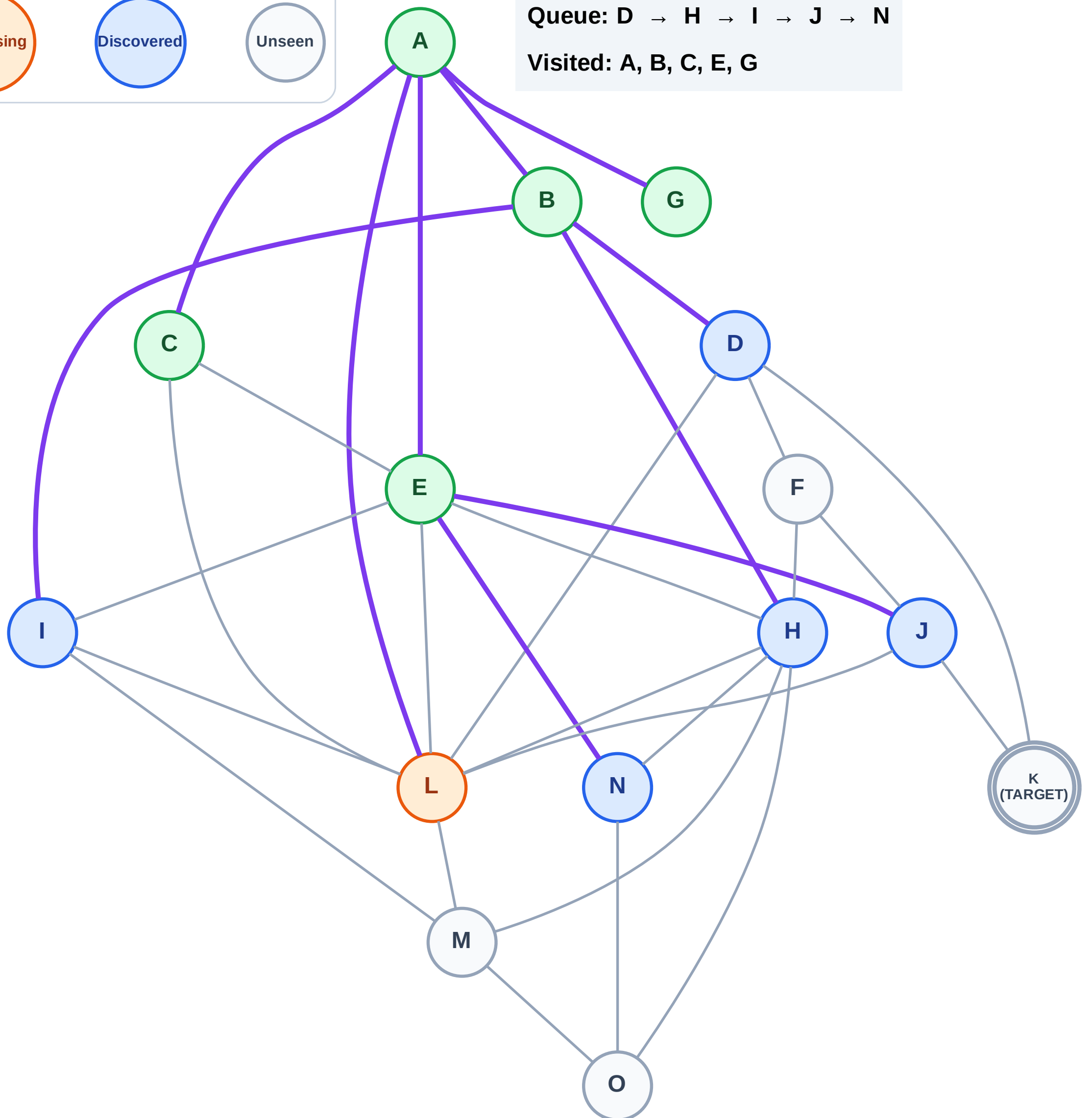
Discovered



Unseen

Queue: D → H → I → J → N

Visited: A, B, C, E, G



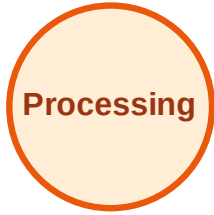
BFS Traversal

Step 13: Discover M from L

Legend



Complete



Processing



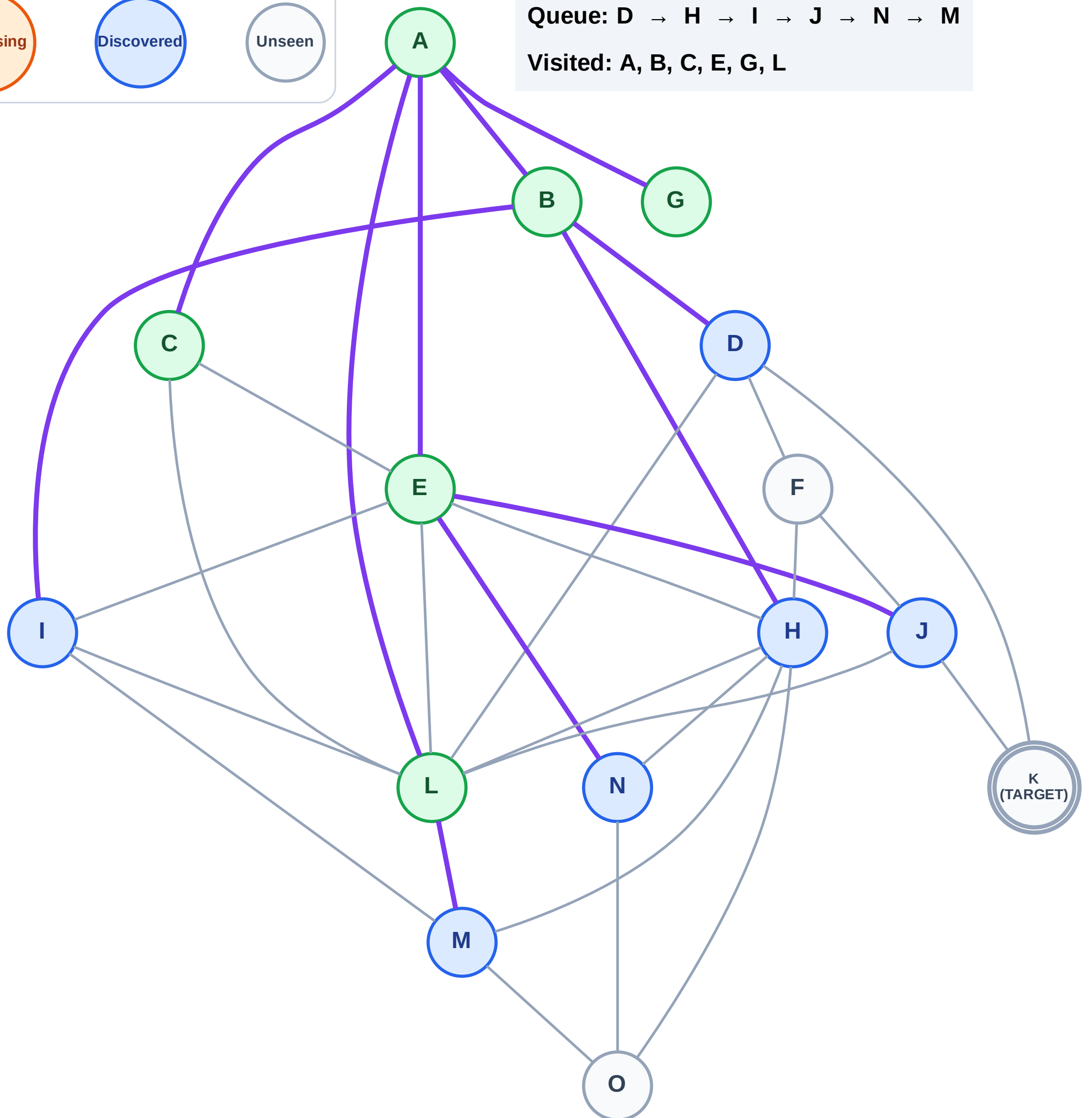
Discovered



Unseen

Queue: D → H → I → J → N → M

Visited: A, B, C, E, G, L



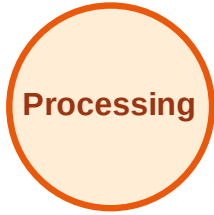
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



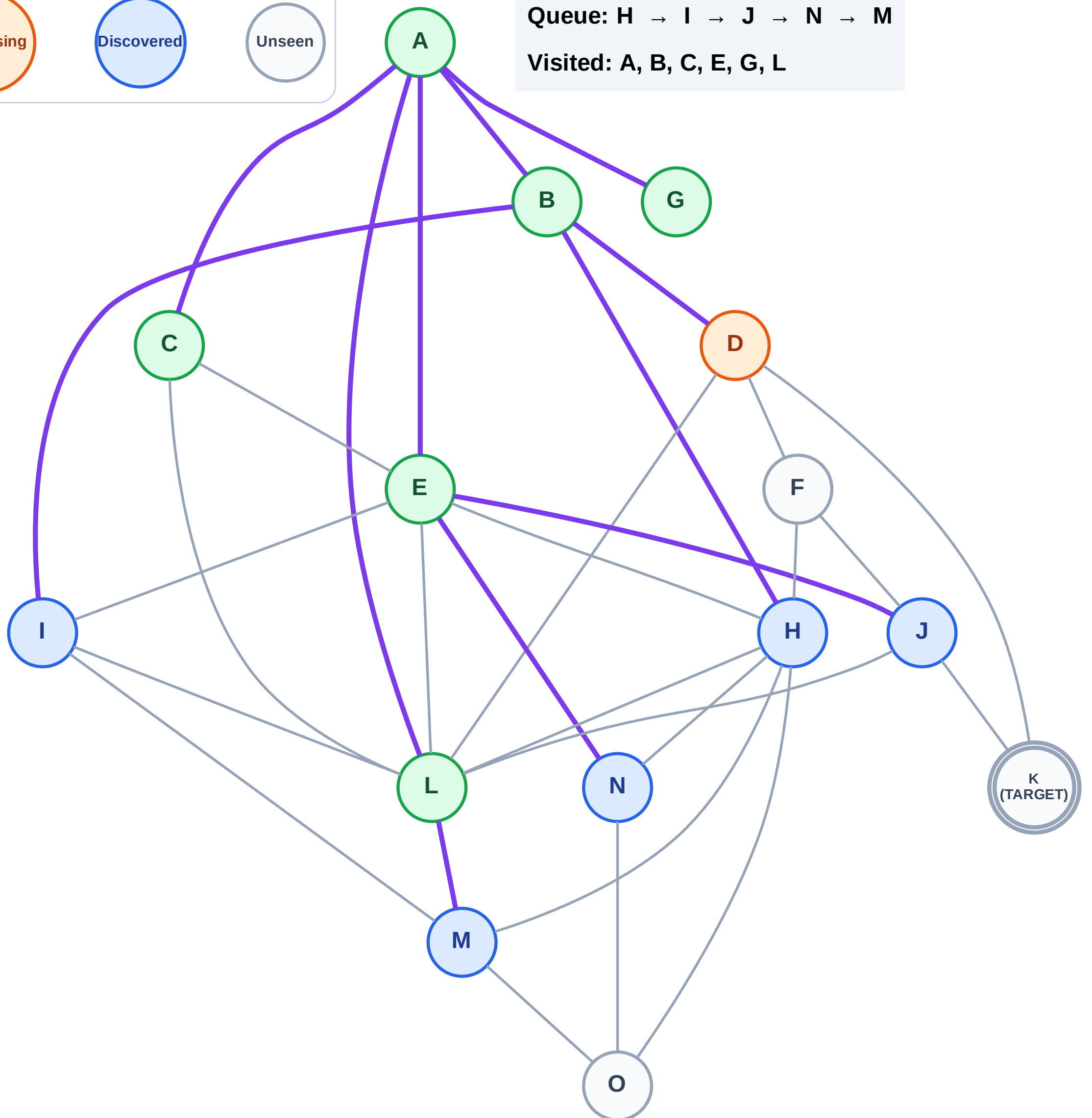
Discovered



Unseen

Queue: H → I → J → N → M

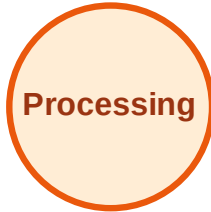
Visited: A, B, C, E, G, L



BFS Traversal

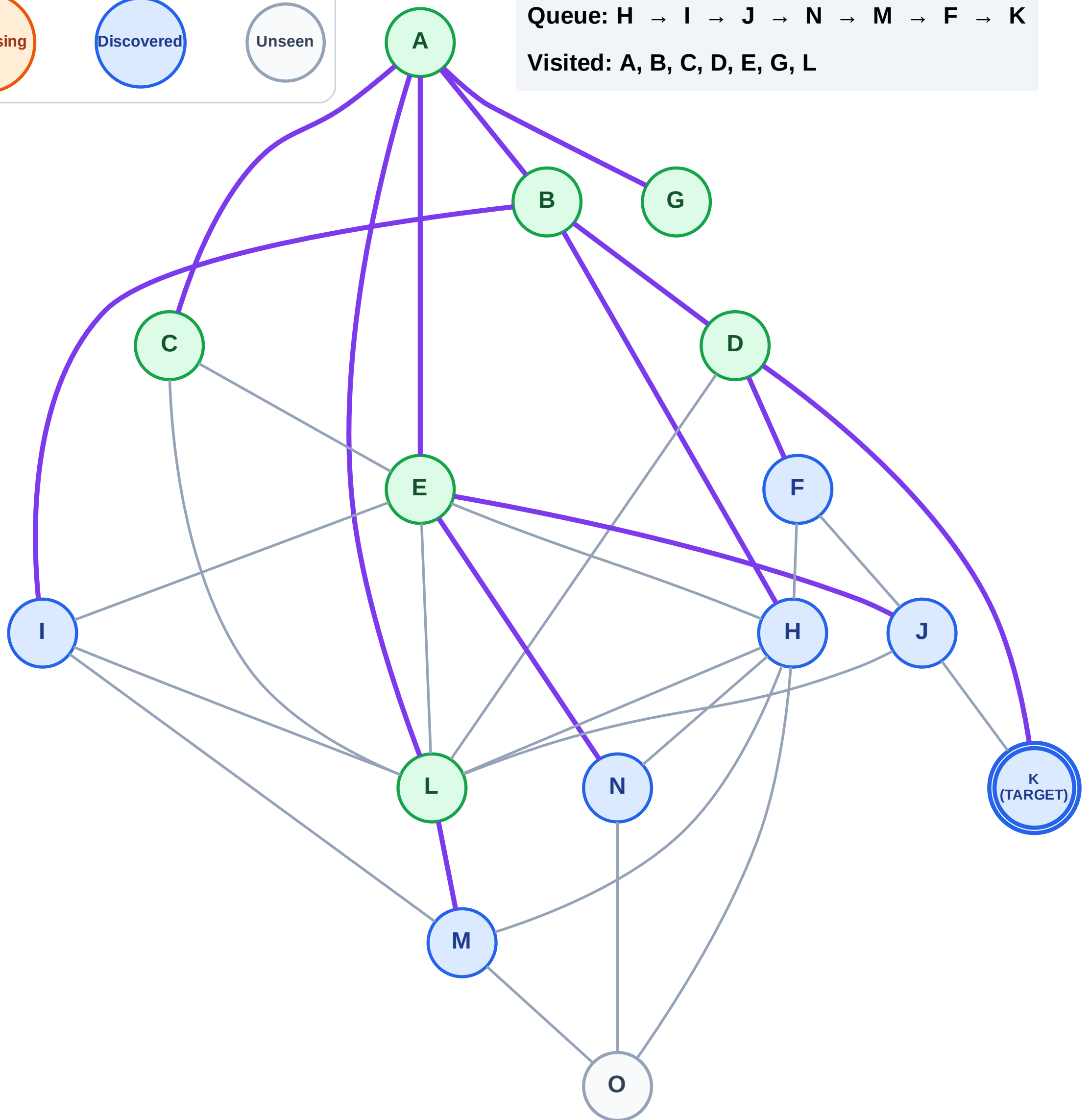
Step 15: Discover F, K from D

Legend



Queue: H → I → J → N → M → F → K

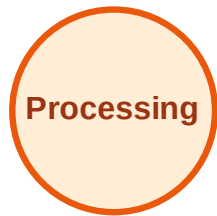
Visited: A, B, C, D, E, G, L



BFS Traversal

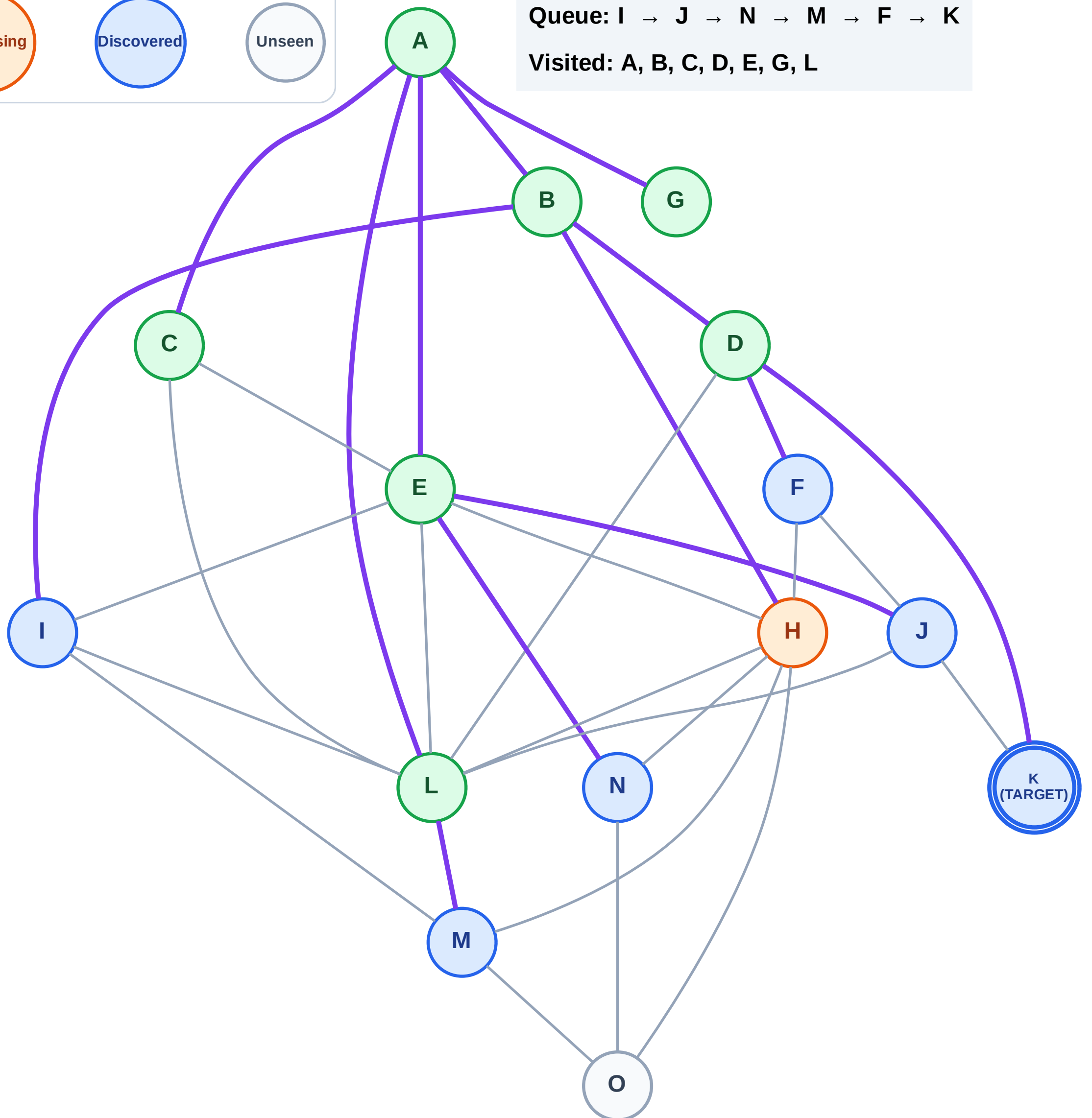
Step 16: Process node H

Legend



Queue: I → J → N → M → F → K

Visited: A, B, C, D, E, G, L



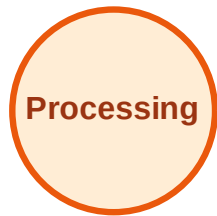
BFS Traversal

Step 17: Discover O from H

Legend



Complete



Processing



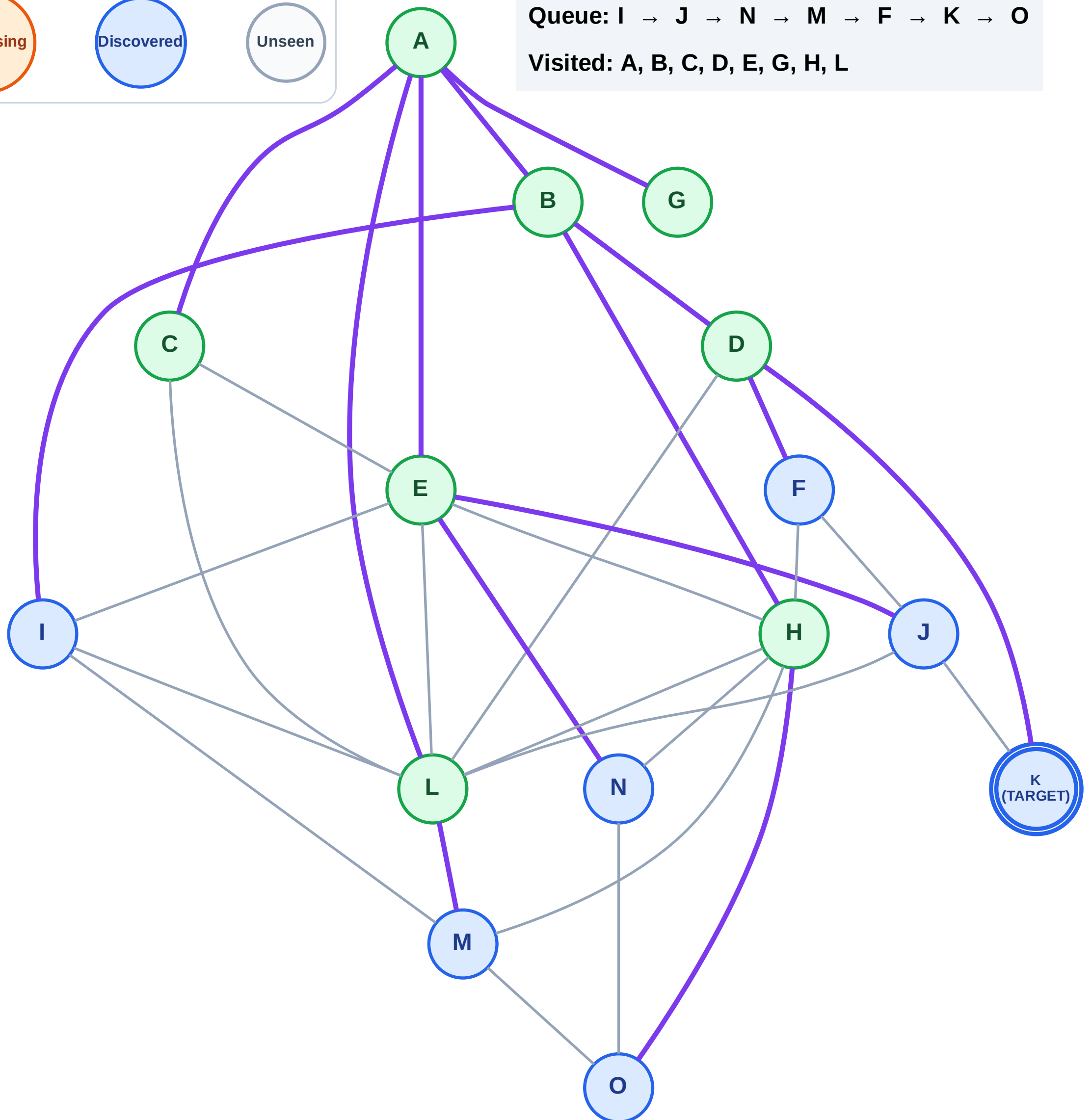
Discovered



Unseen

Queue: I → J → N → M → F → K → O

Visited: A, B, C, D, E, G, H, L



BFS Traversal

Step 18: Process node I

Legend

Complete

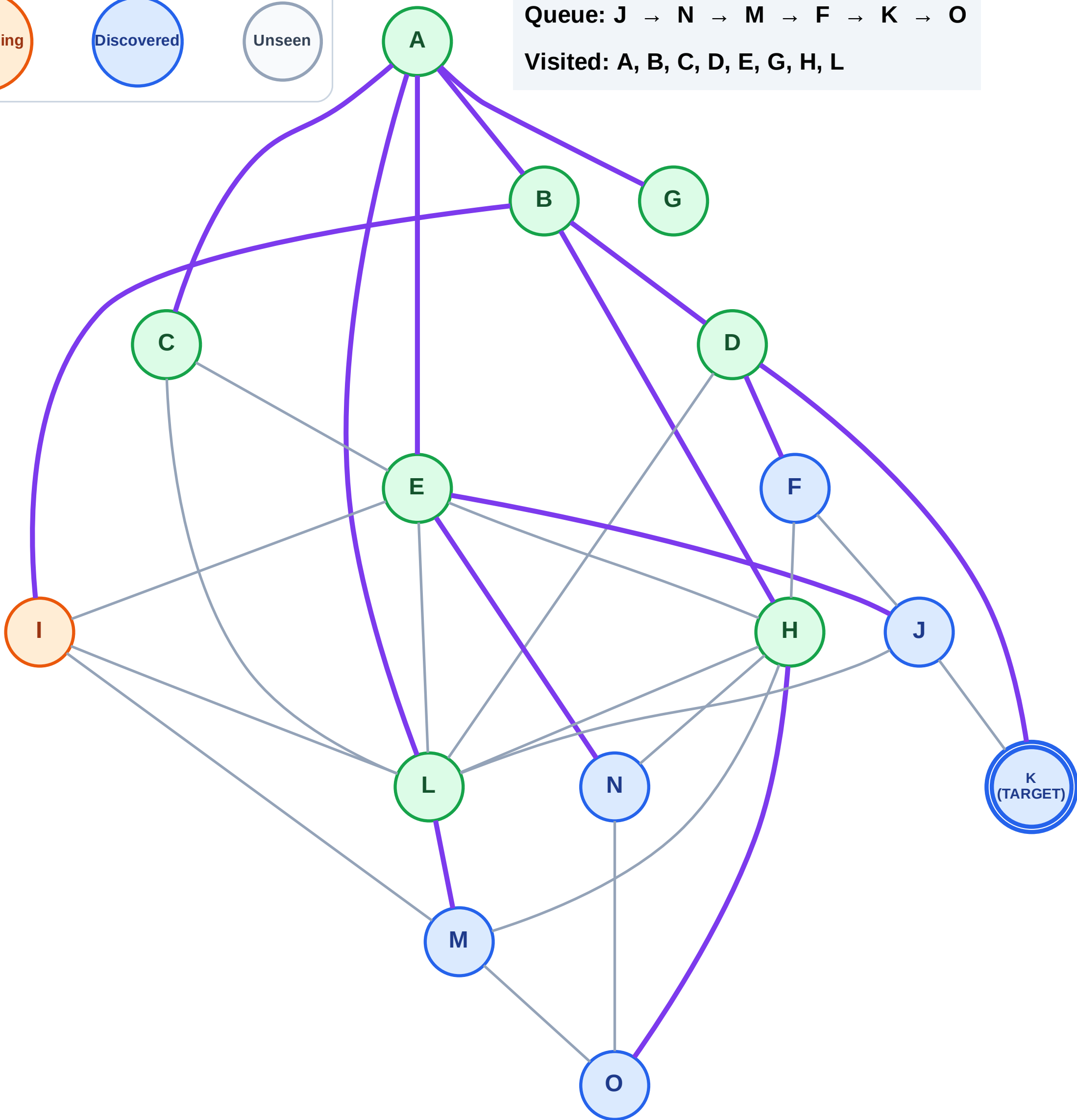
Processing

Discovered

Unseen

Queue: J → N → M → F → K → O

Visited: A, B, C, D, E, G, H, L



BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



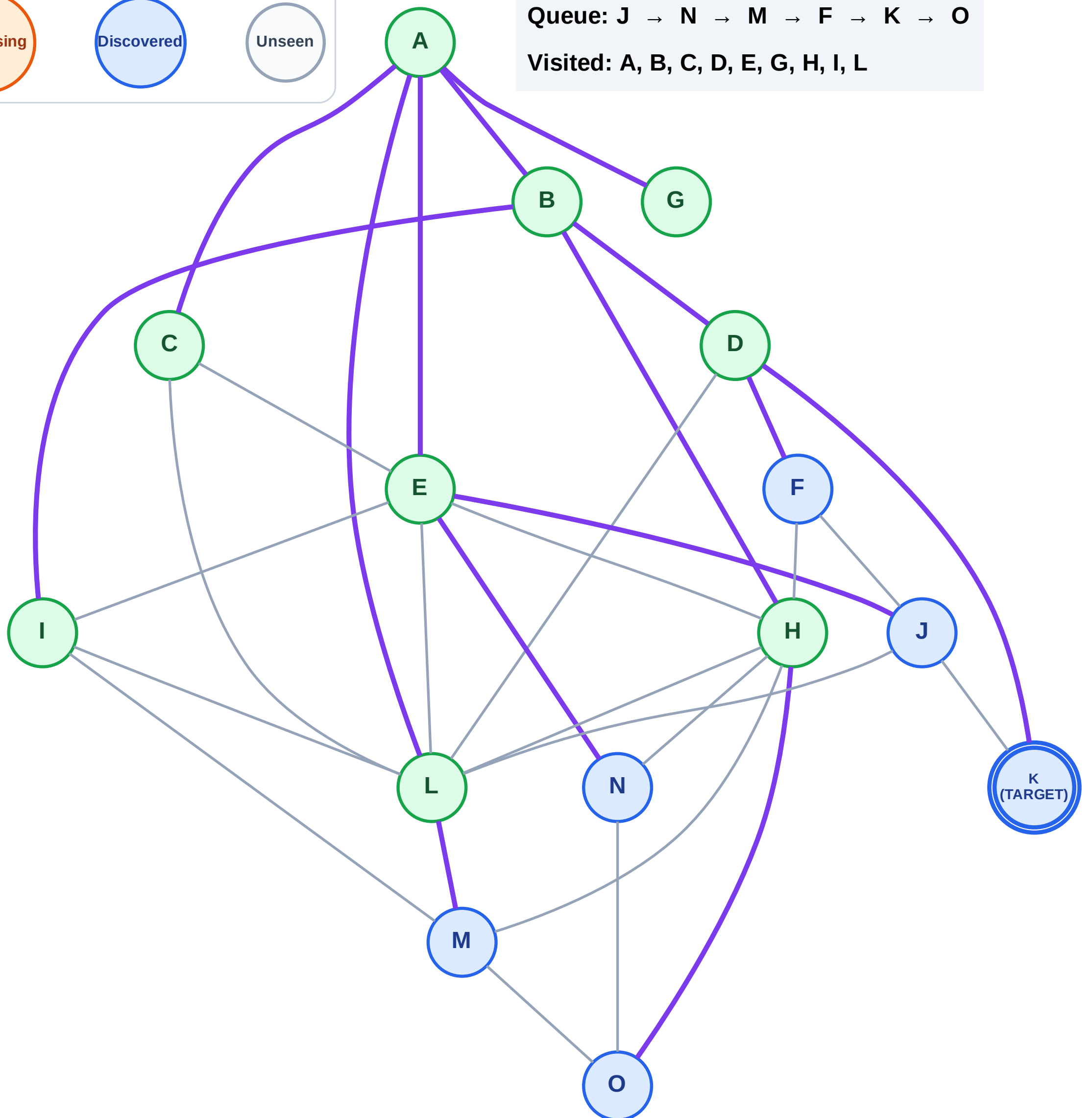
Discovered



Unseen

Queue: J → N → M → F → K → O

Visited: A, B, C, D, E, G, H, I, L



BFS Traversal

Step 20: Process node J

Legend



Complete



Processing



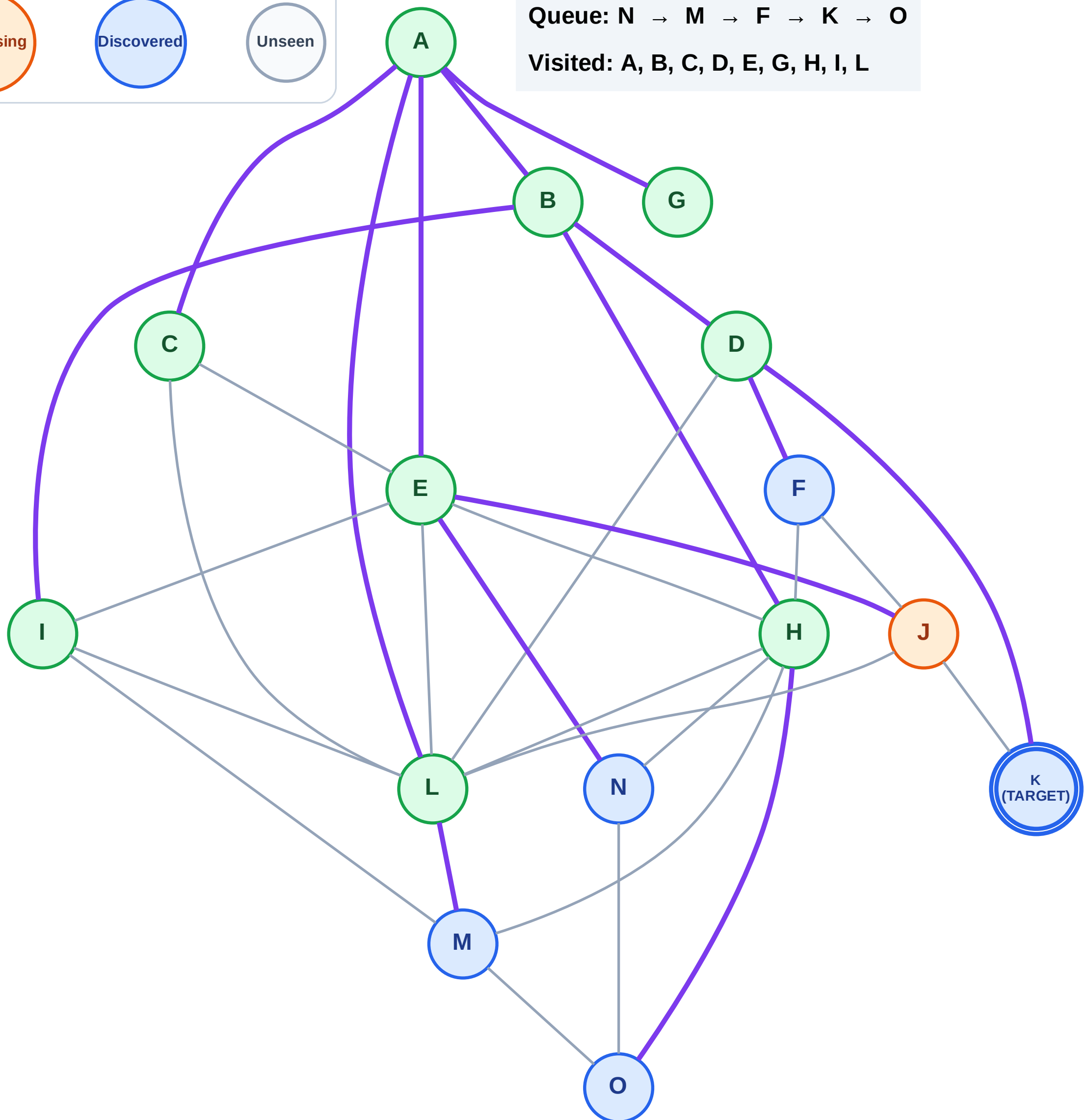
Discovered



Unseen

Queue: N → M → F → K → O

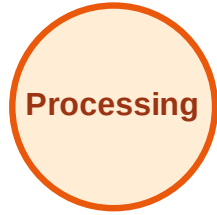
Visited: A, B, C, D, E, G, H, I, L



BFS Traversal

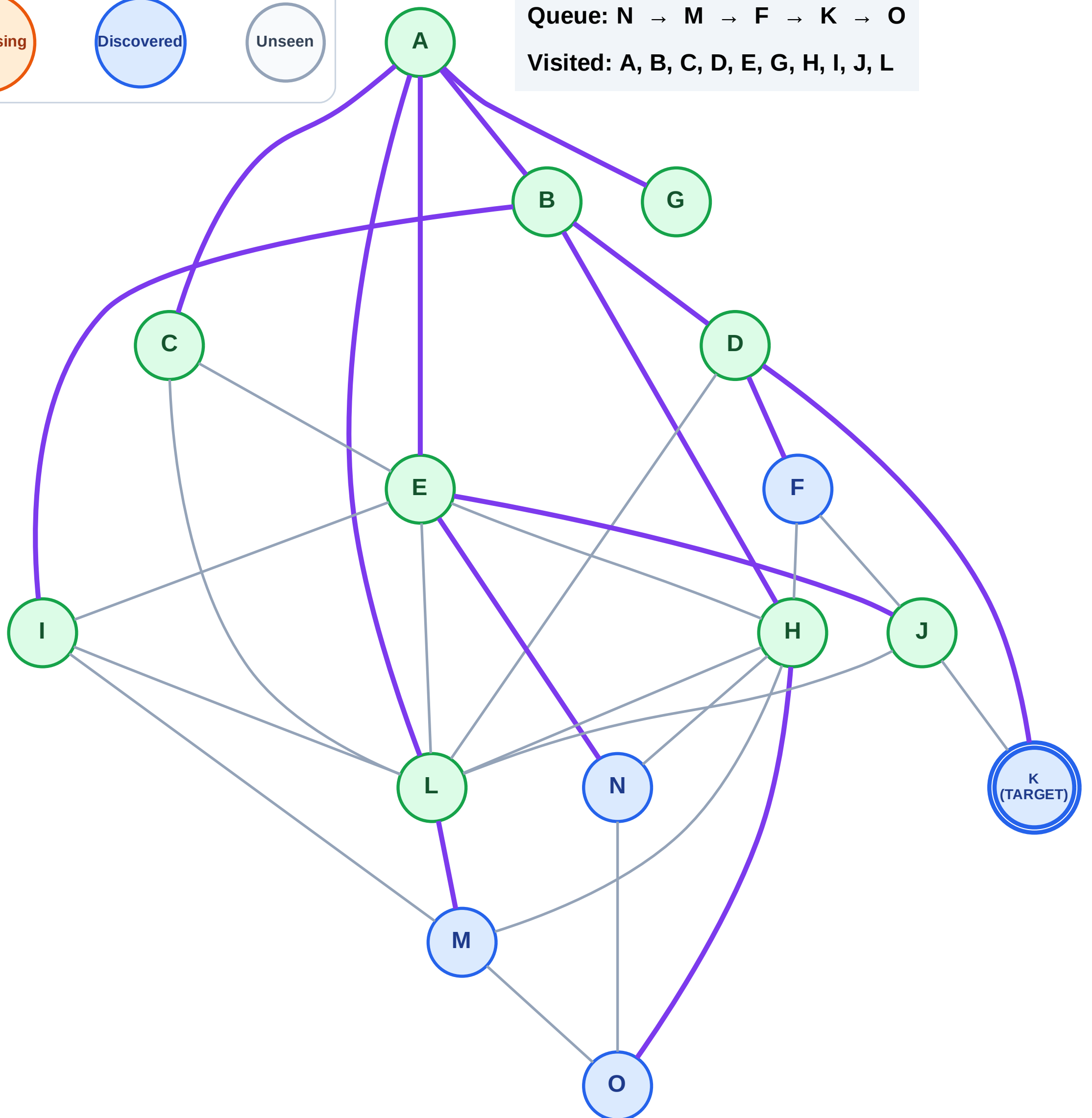
Step 21: No new neighbors from J

Legend



Queue: N → M → F → K → O

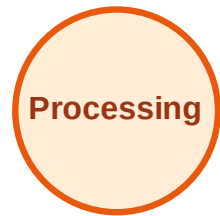
Visited: A, B, C, D, E, G, H, I, J, L



BFS Traversal

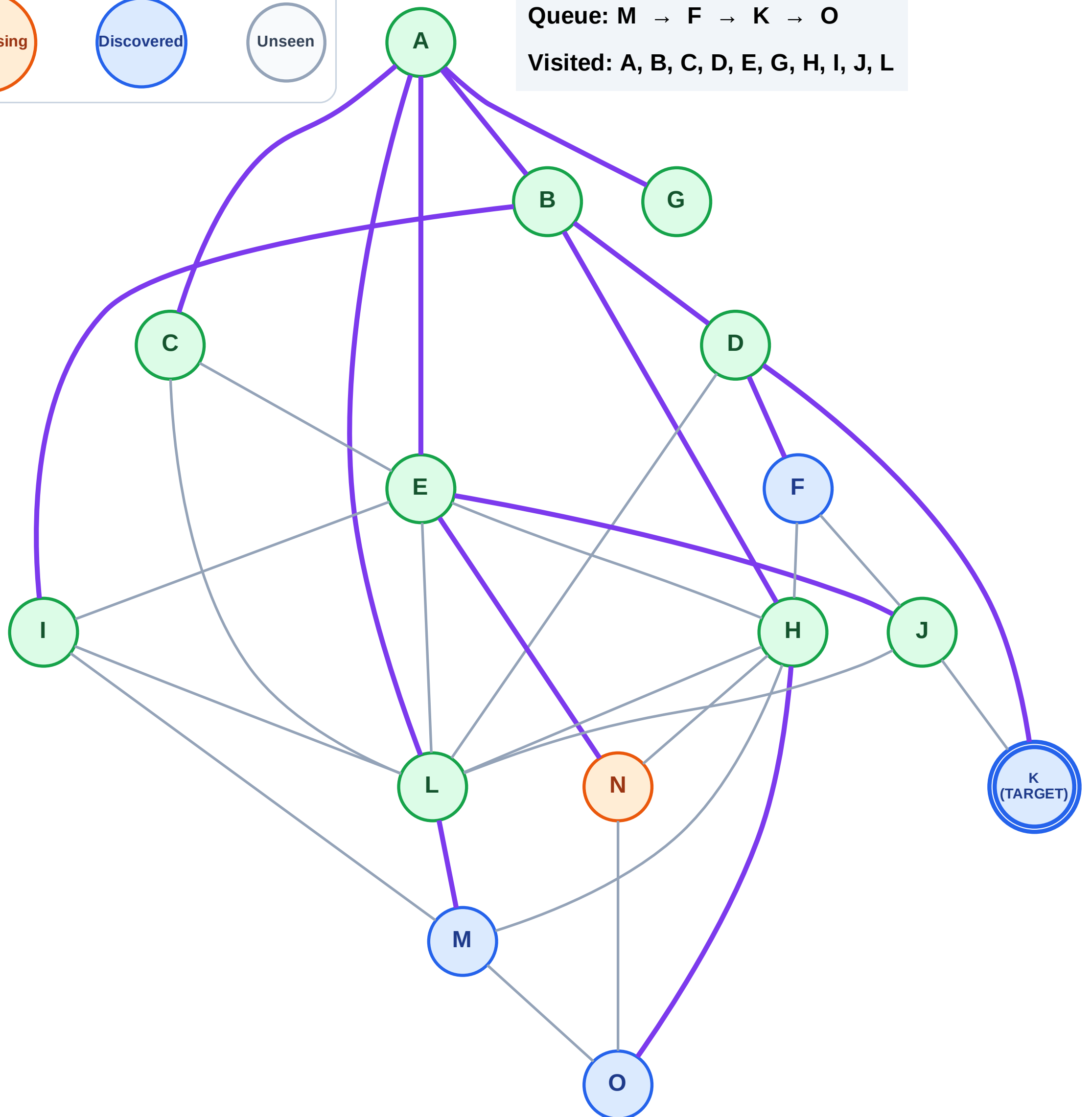
Step 22: Process node N

Legend



Queue: M → F → K → O

Visited: A, B, C, D, E, G, H, I, J, L



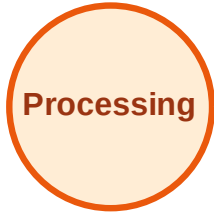
BFS Traversal

Step 23: No new neighbors from N

Legend



Complete



Processing



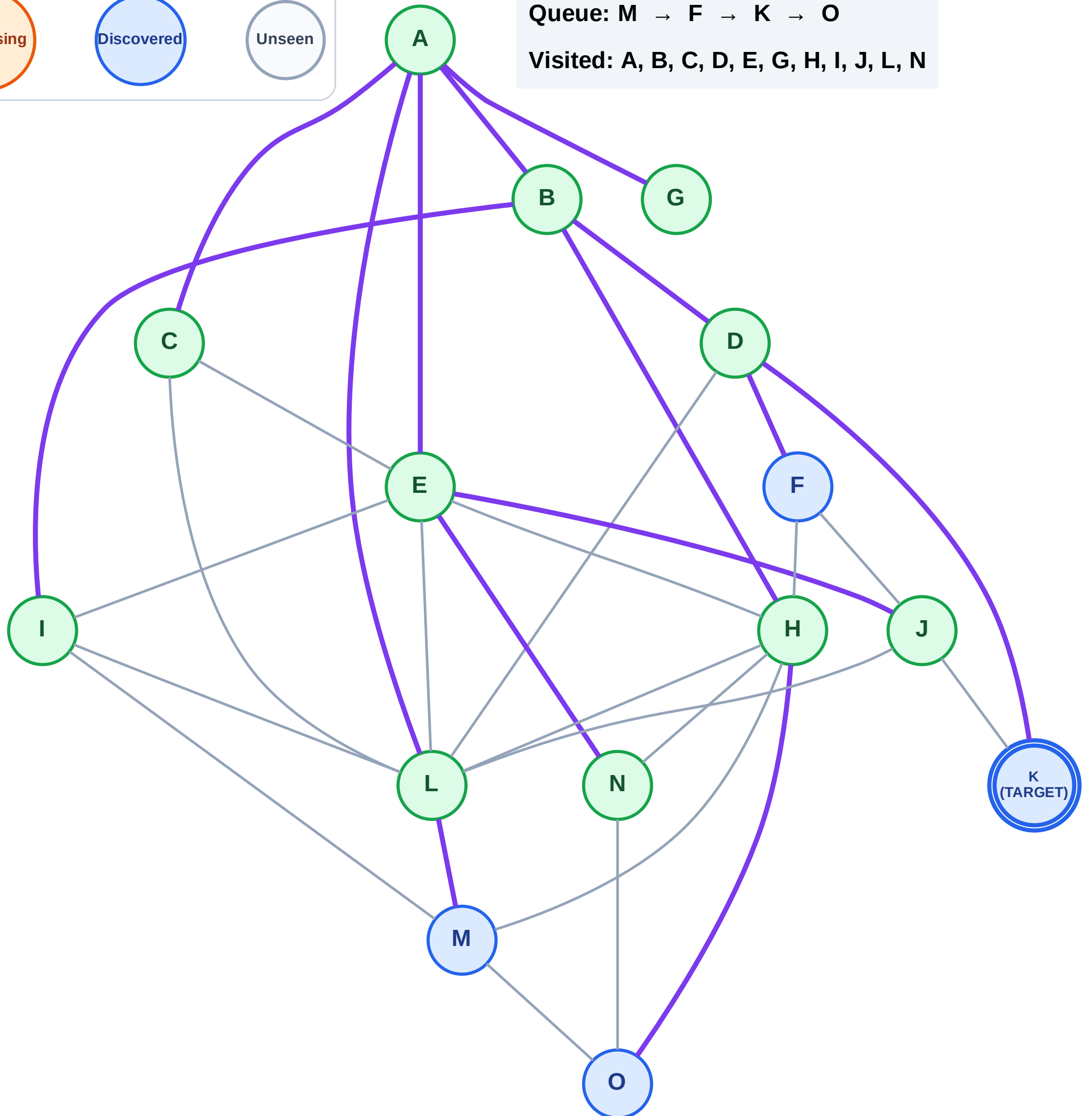
Discovered



Unseen

Queue: M → F → K → O

Visited: A, B, C, D, E, G, H, I, J, L, N



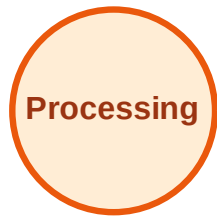
BFS Traversal

Step 24: Process node M

Legend



Complete



Processing



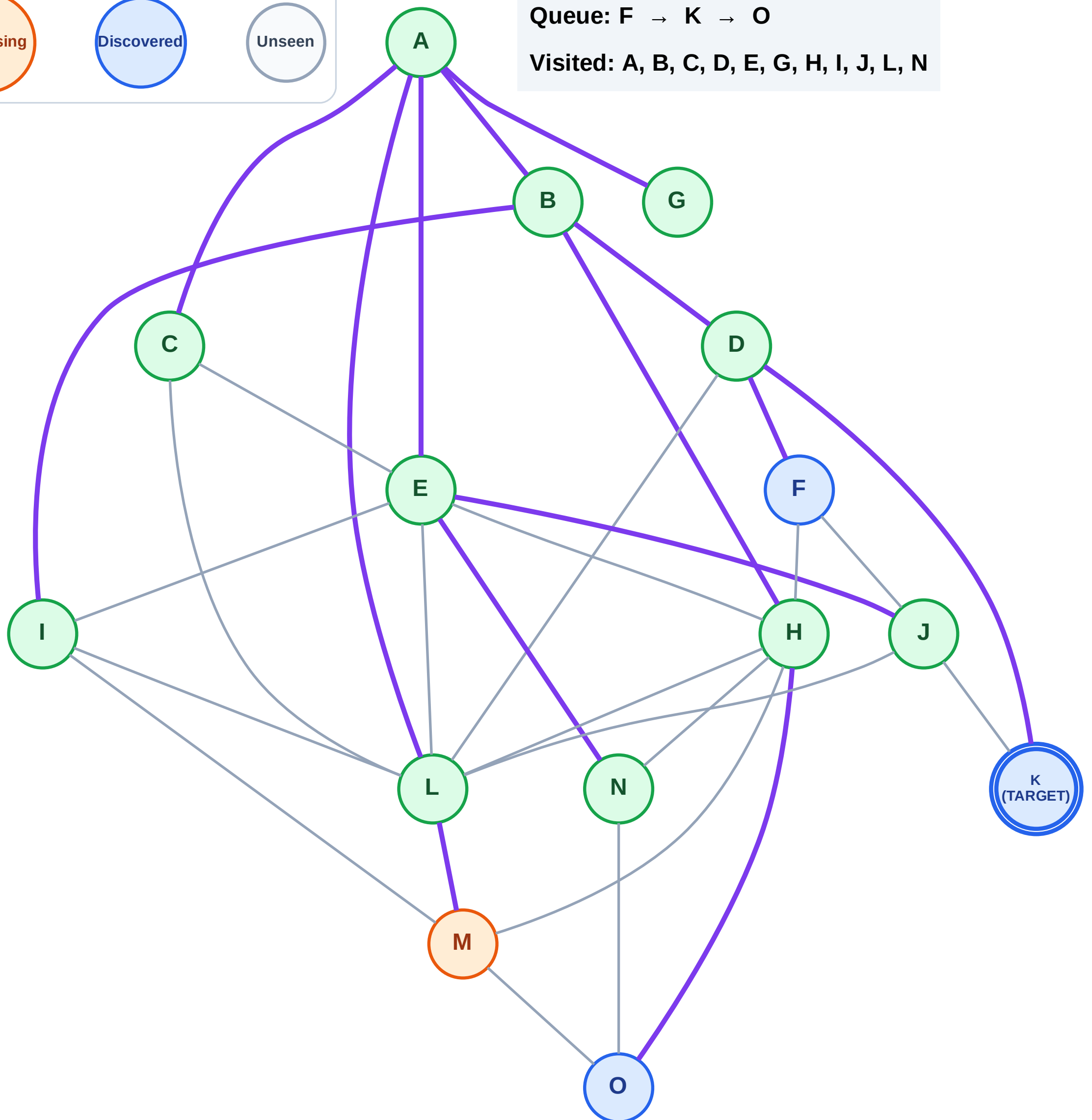
Discovered



Unseen

Queue: F → K → O

Visited: A, B, C, D, E, G, H, I, J, L, N



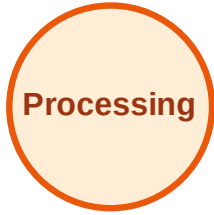
BFS Traversal

Step 25: No new neighbors from M

Legend



Complete



Processing



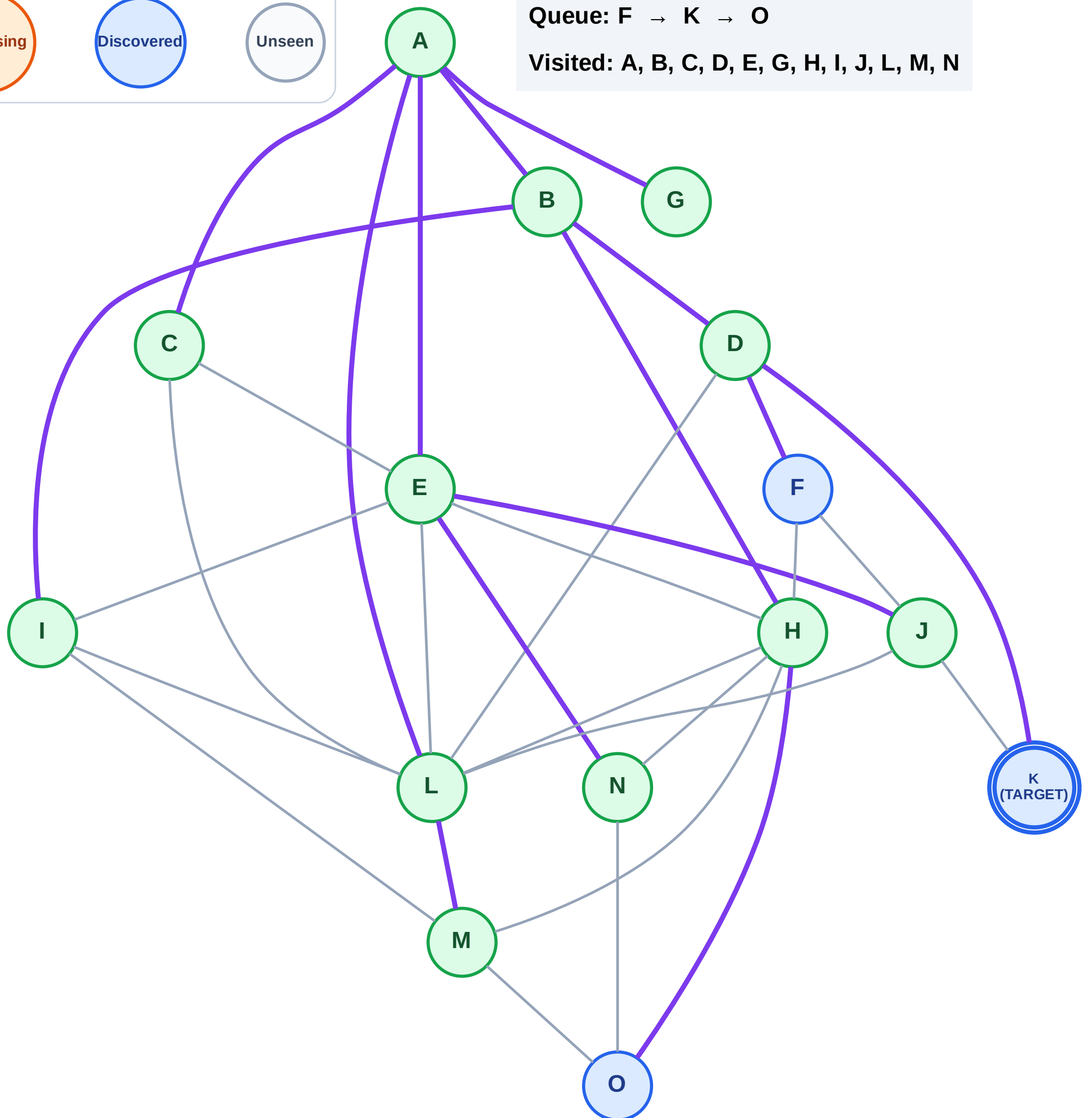
Discovered



Unseen

Queue: F → K → O

Visited: A, B, C, D, E, G, H, I, J, L, M, N



BFS Traversal

Step 26: Process node F

Legend

Complete

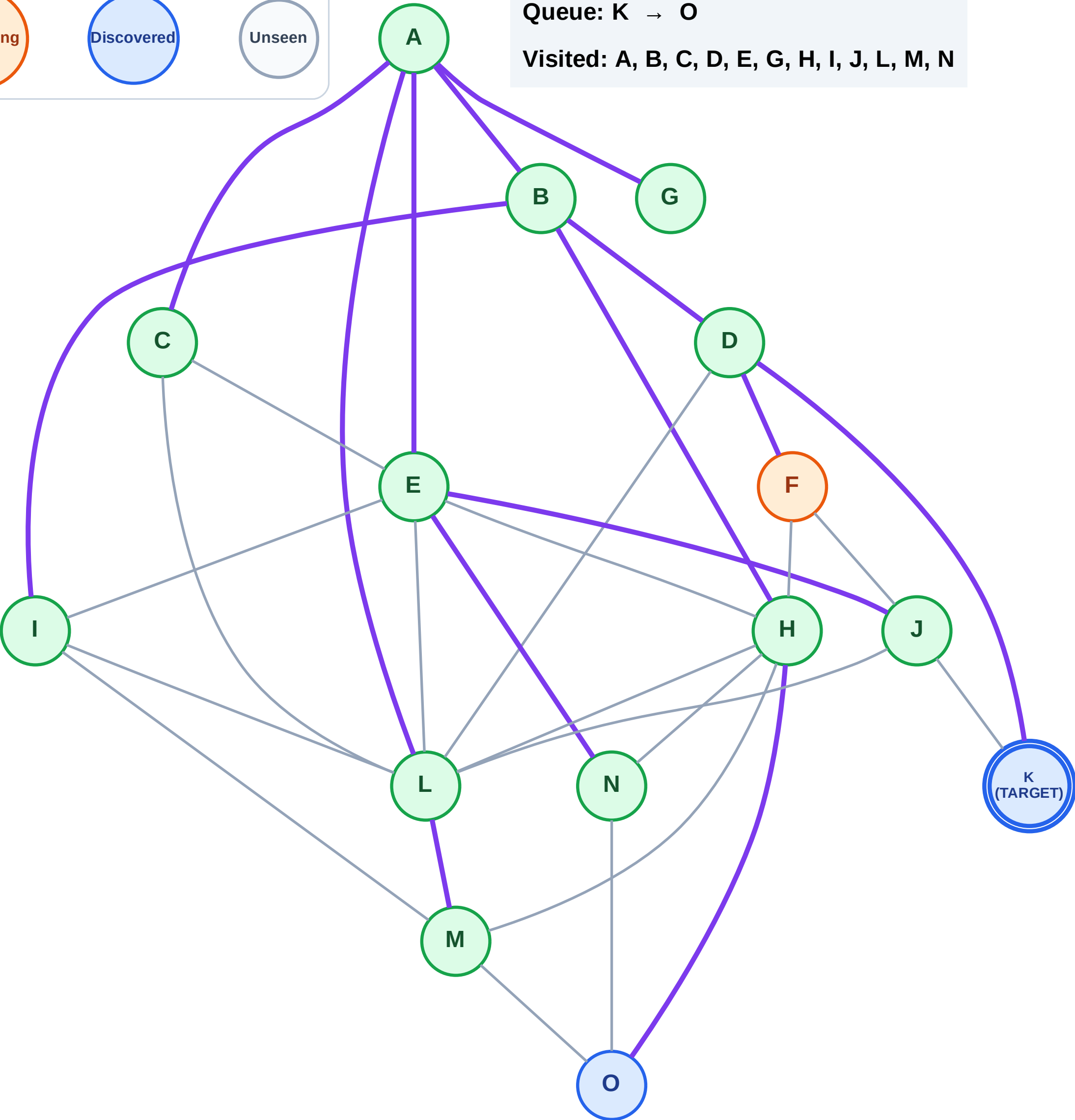
Processing

Discovered

Unseen

Queue: K → O

Visited: A, B, C, D, E, G, H, I, J, L, M, N



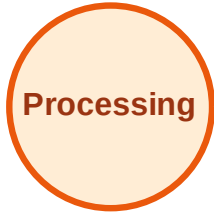
BFS Traversal

Step 27: No new neighbors from F

Legend



Complete



Processing



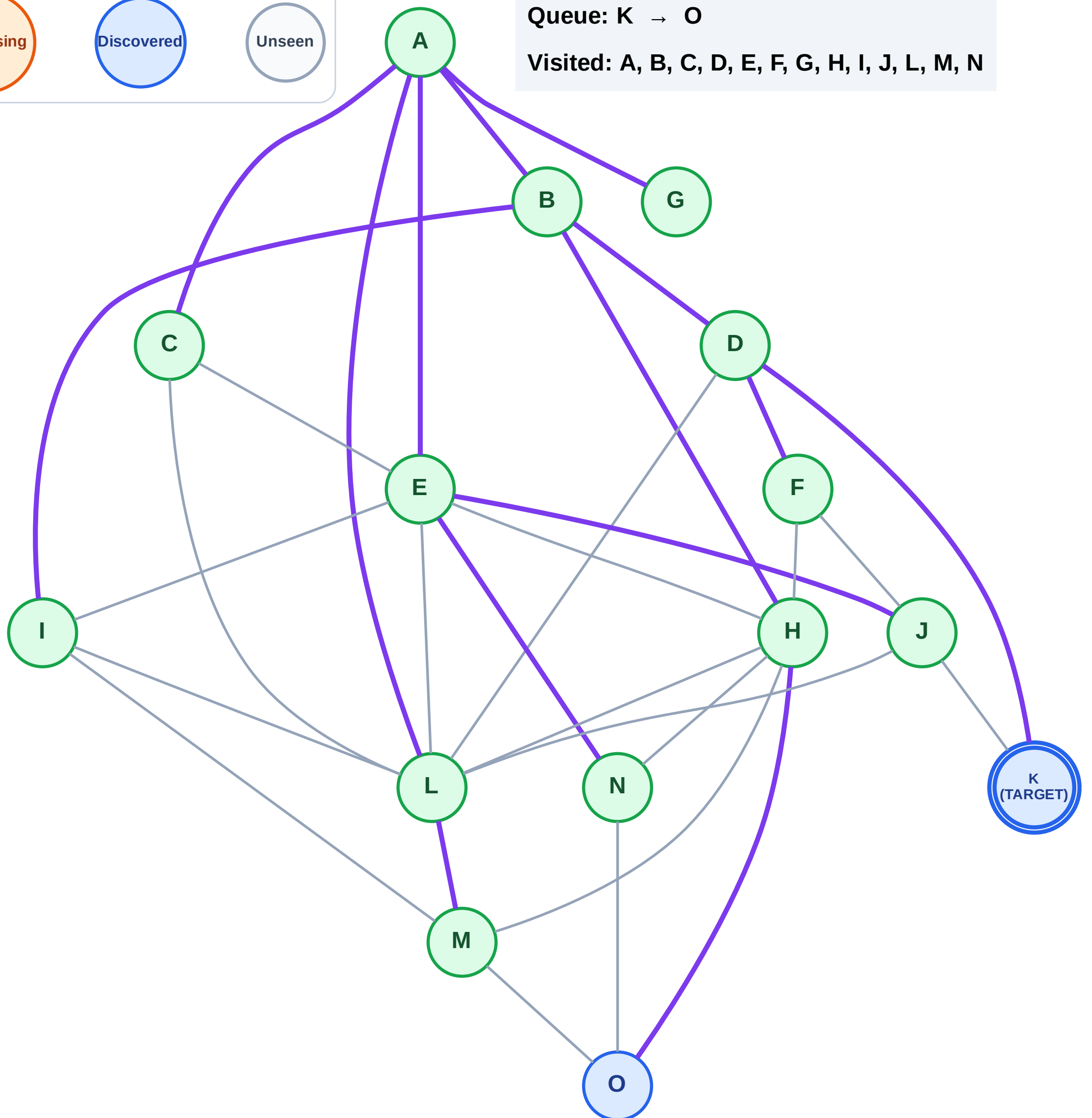
Discovered



Unseen

Queue: K → O

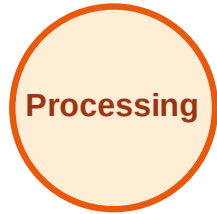
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

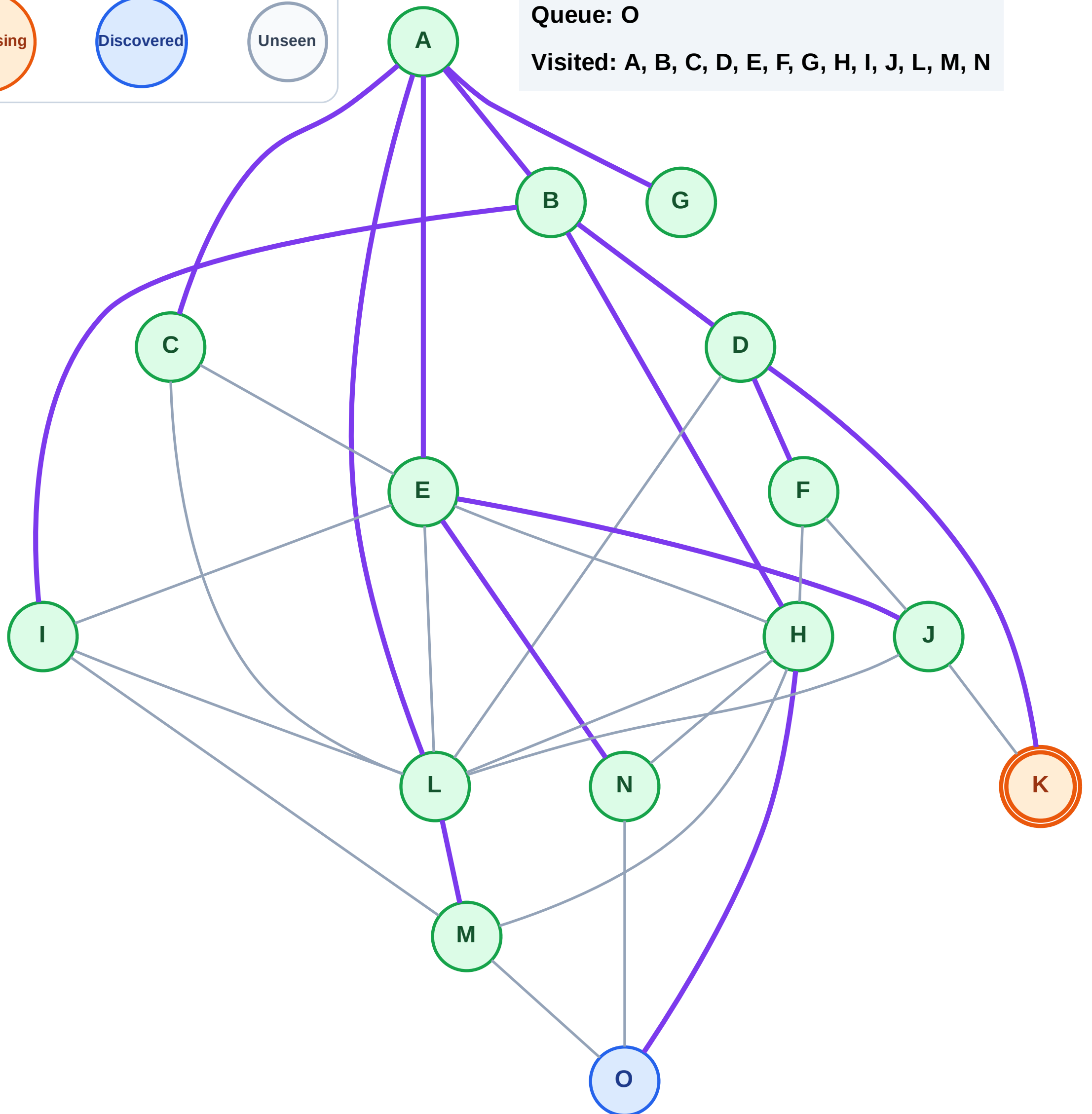
Step 28: Process node K

Legend



Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

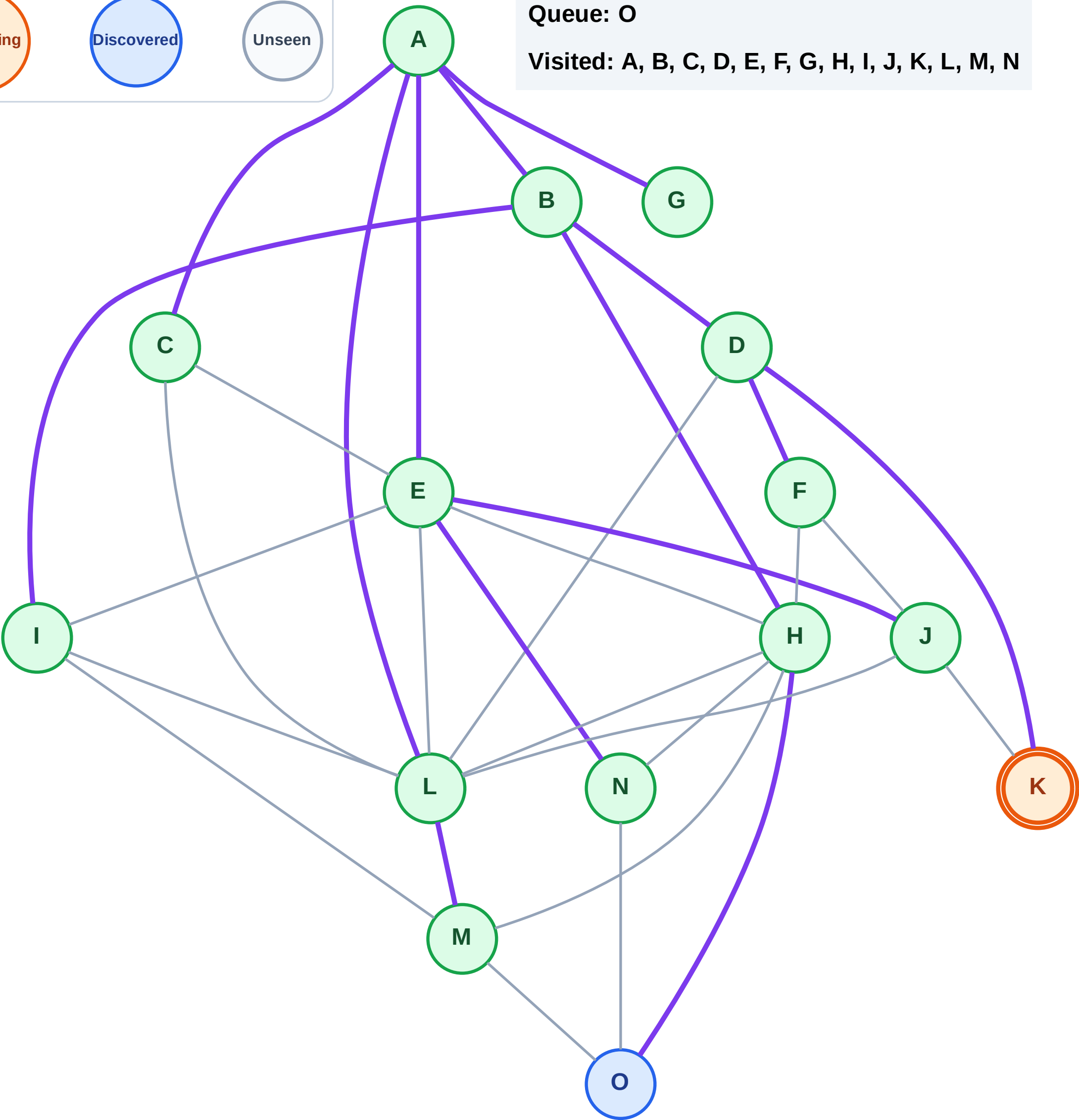
Processing

Discovered

Unseen

Queue: O

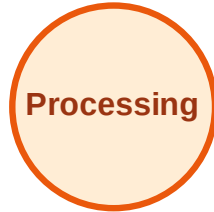
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

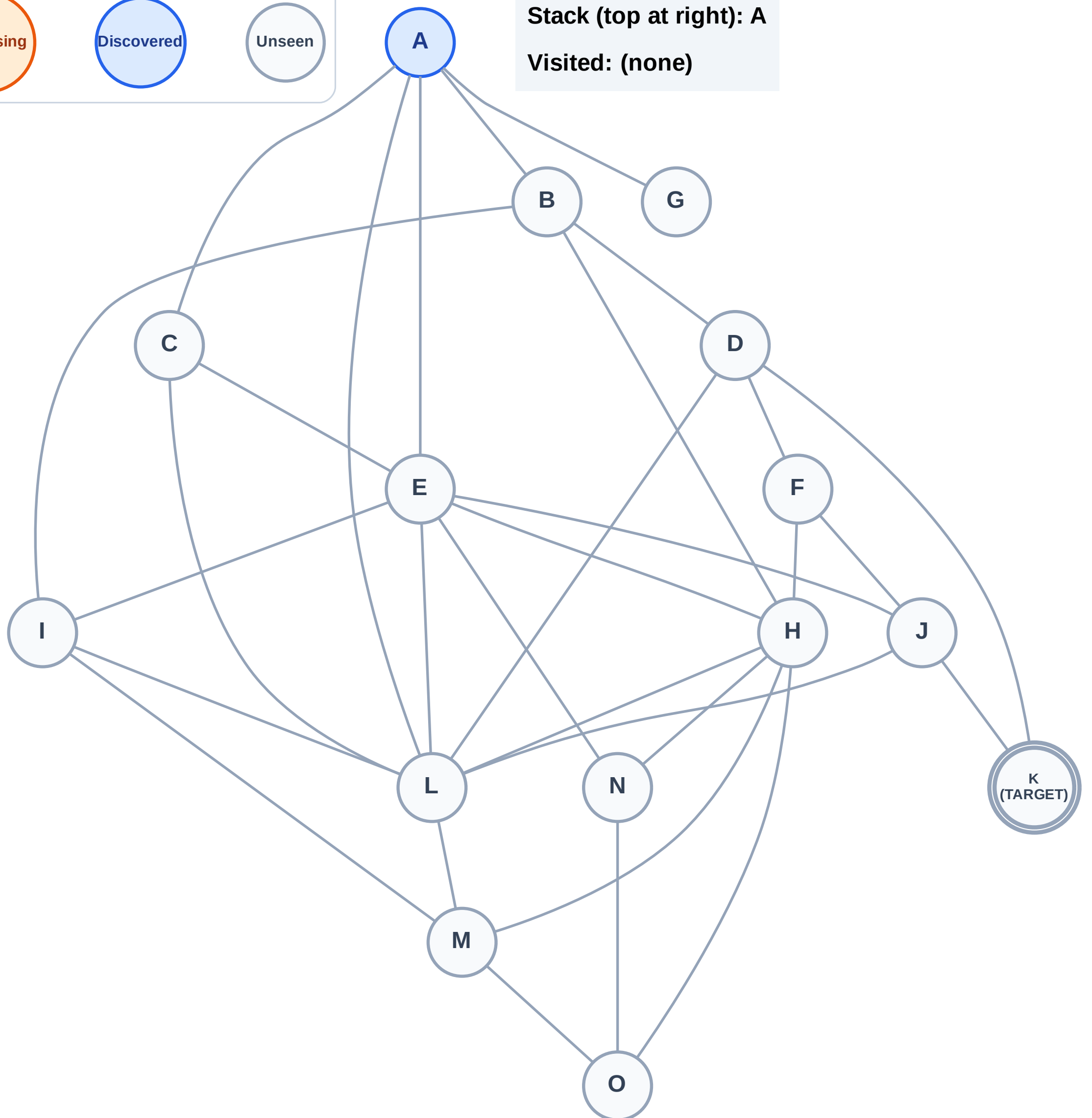
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

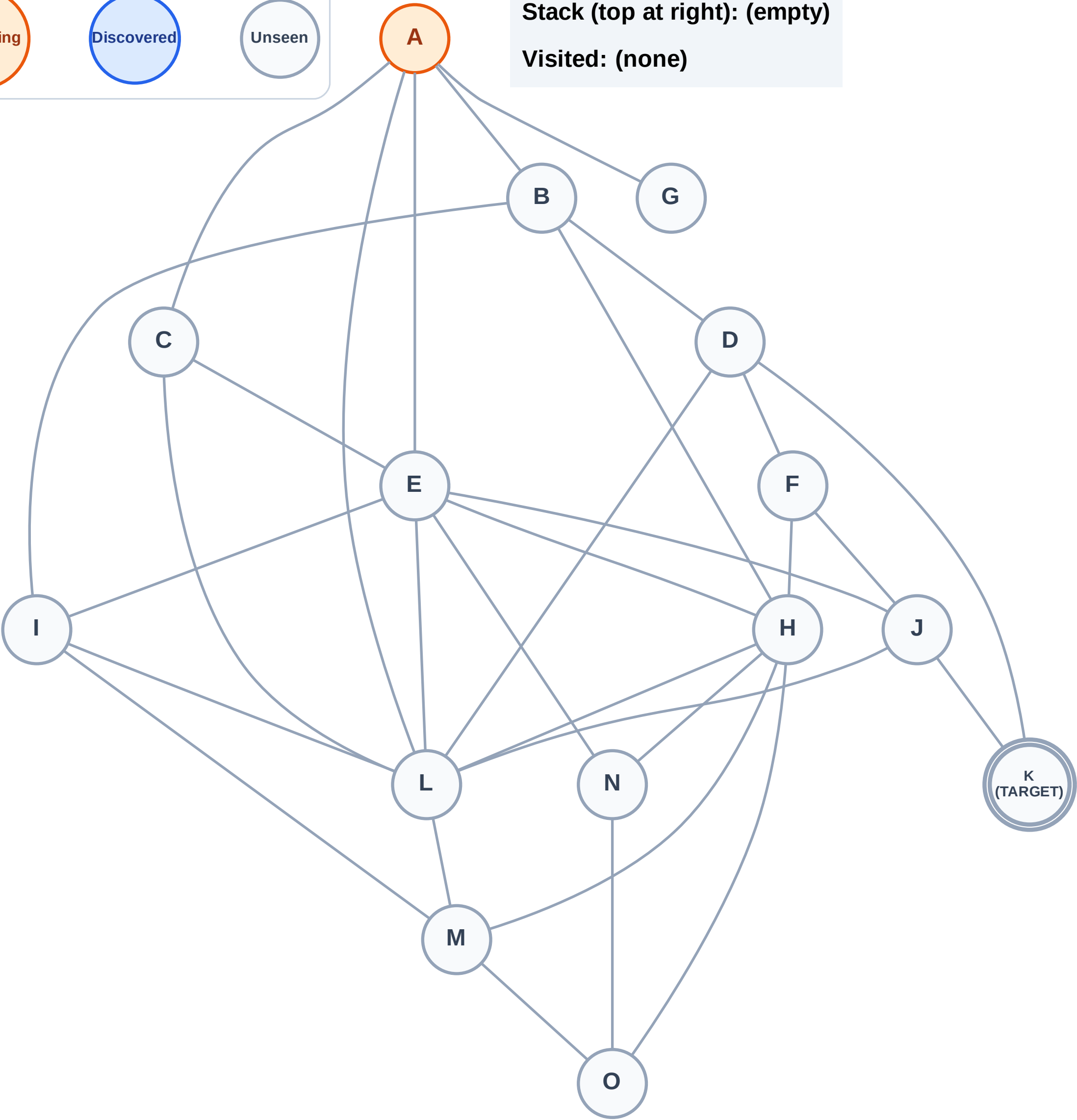
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



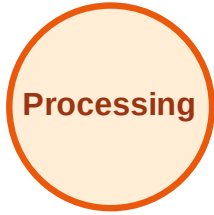
DFS Traversal

Step 3: Discover L, G, E, C, B from A

Legend



Complete



Processing



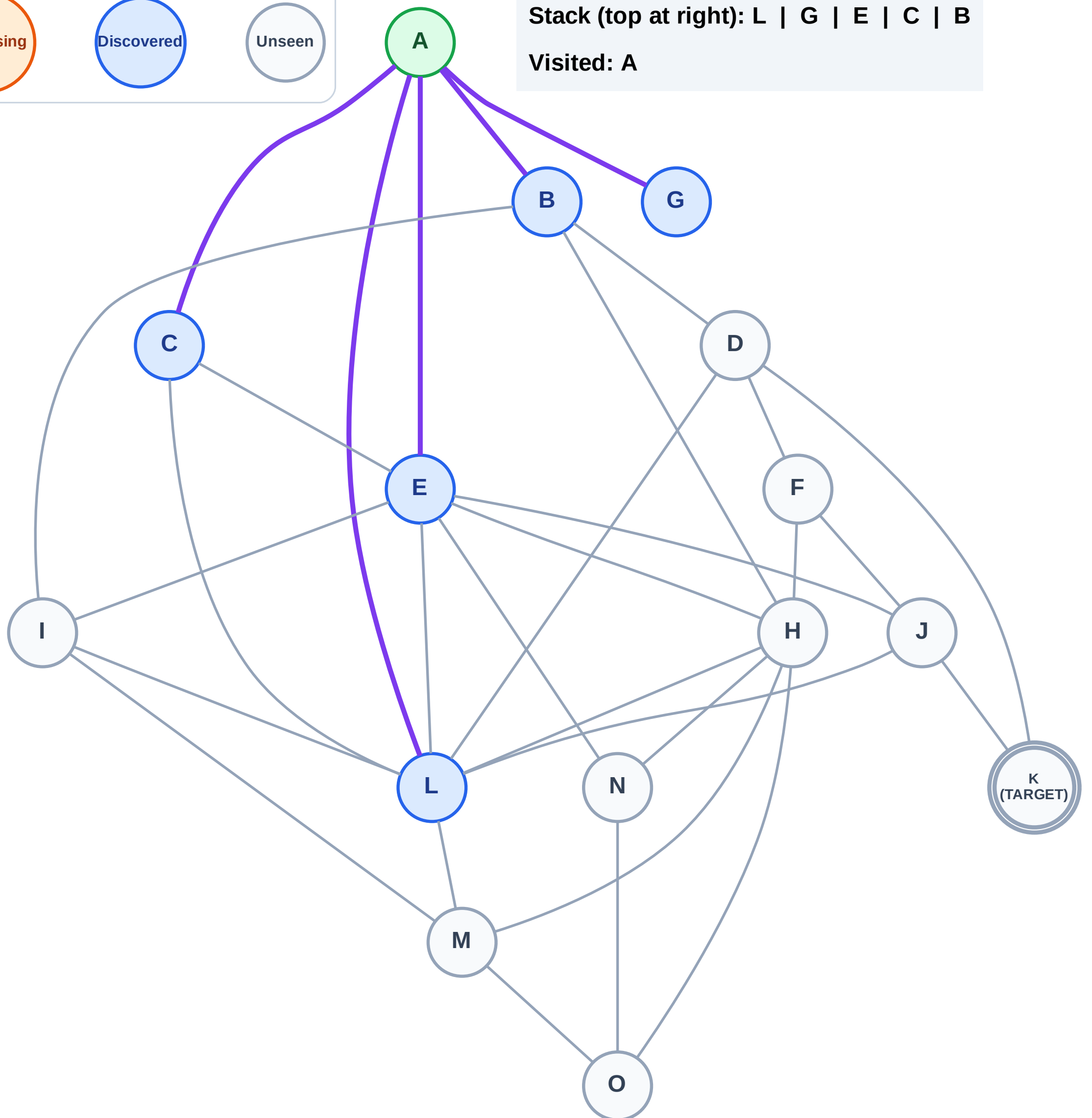
Discovered



Unseen

Stack (top at right): L | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

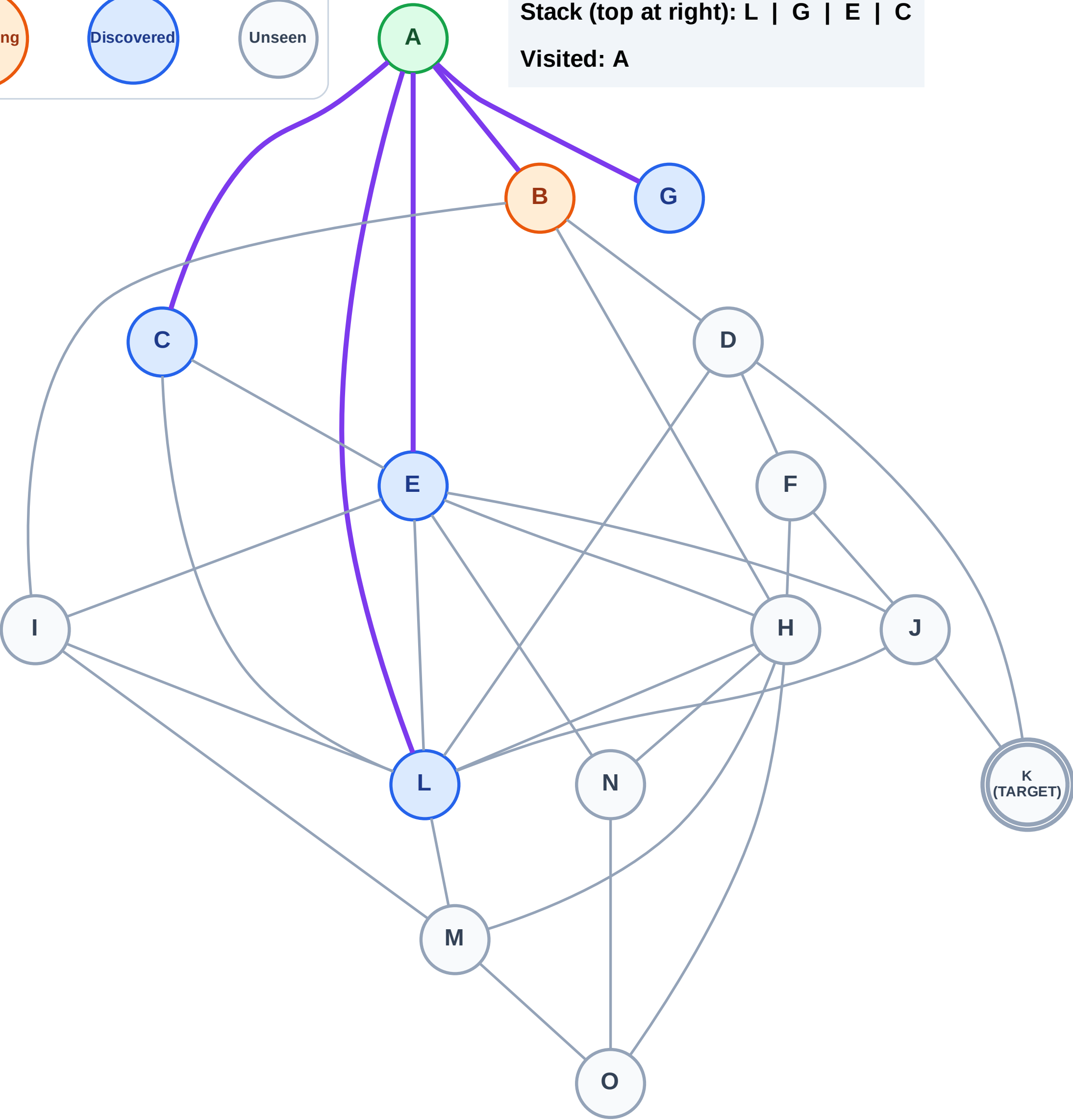
Processing

Discovered

Unseen

Stack (top at right): L | G | E | C

Visited: A



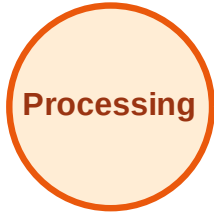
DFS Traversal

Step 5: Discover I, H, D from B

Legend



Complete



Processing



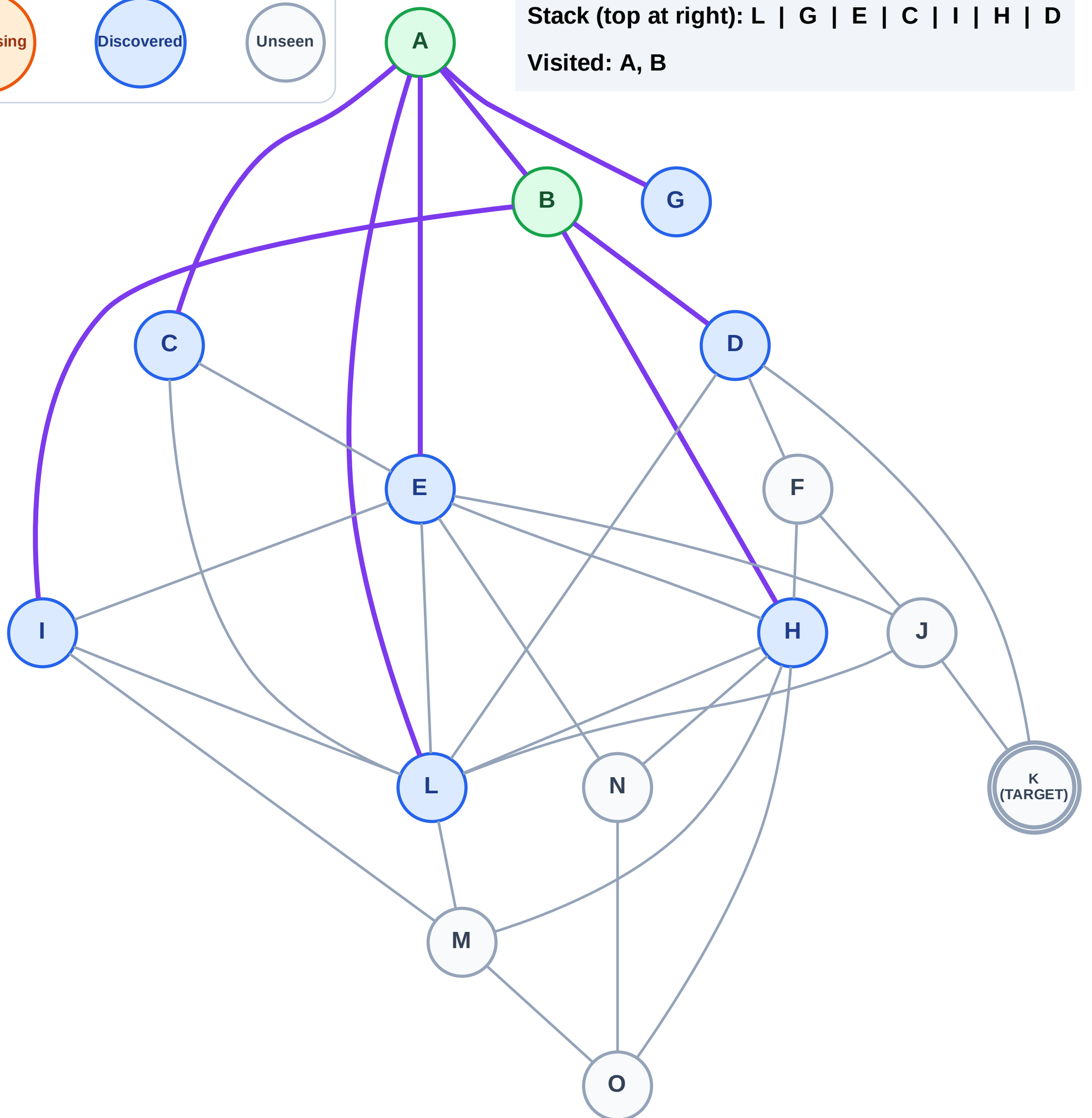
Discovered



Unseen

Stack (top at right): L | G | E | C | I | H | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

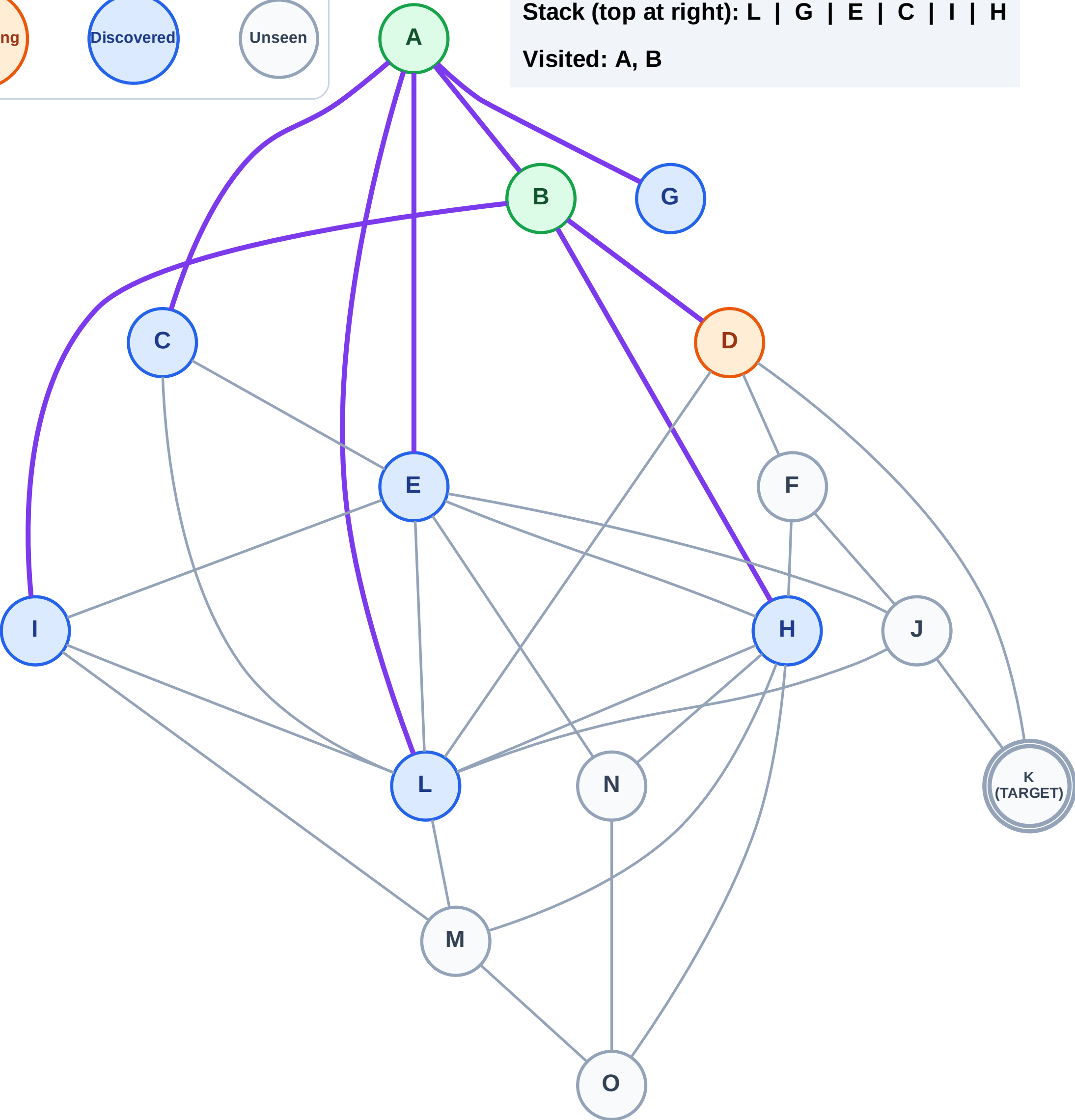
Processing

Discovered

Unseen

Stack (top at right): L | G | E | C | I | H

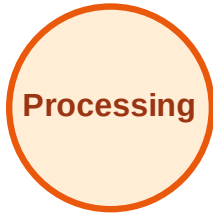
Visited: A, B



DFS Traversal

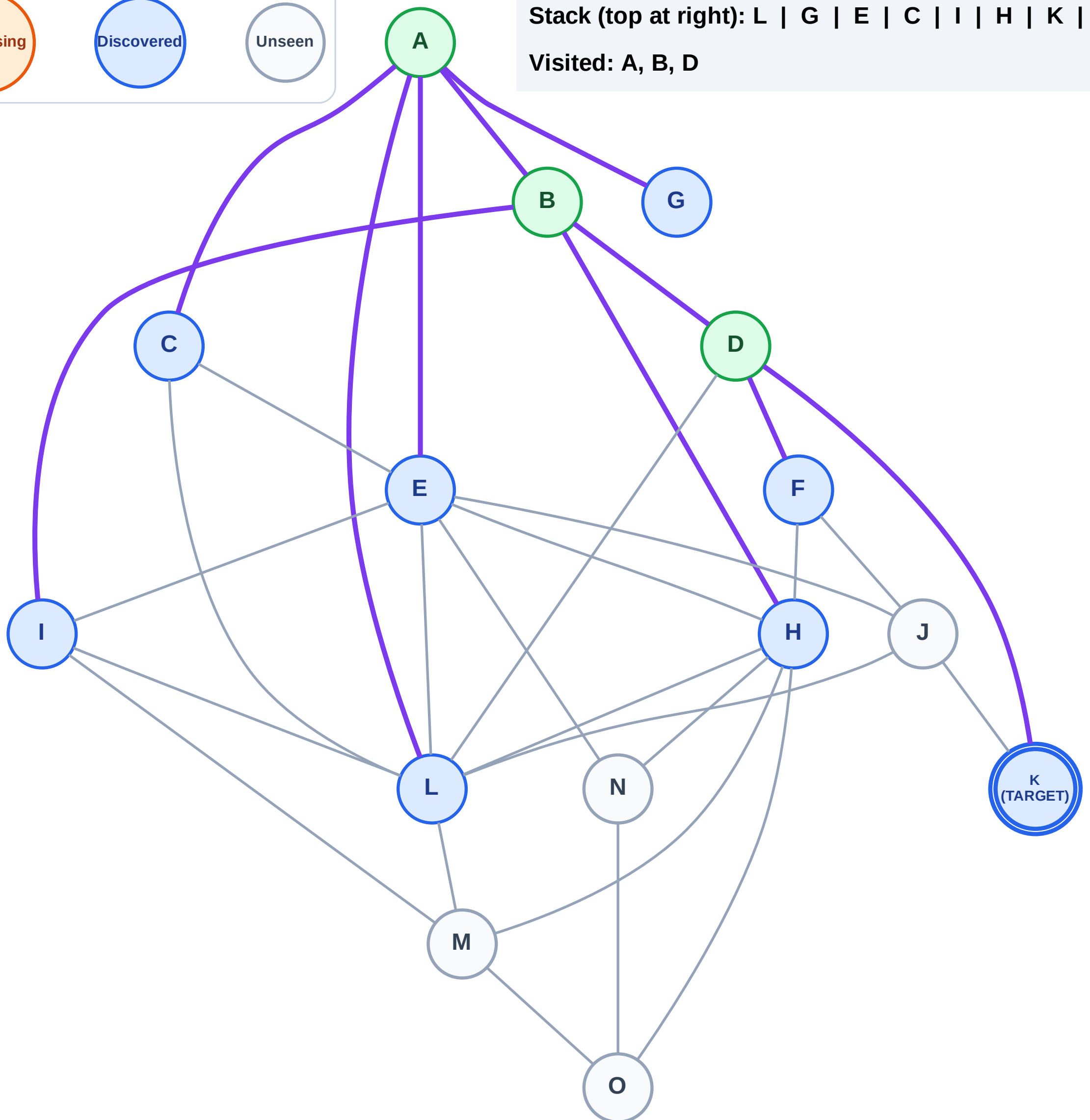
Step 7: Discover K, F from D

Legend



Stack (top at right): L | G | E | C | I | H | K | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

Complete

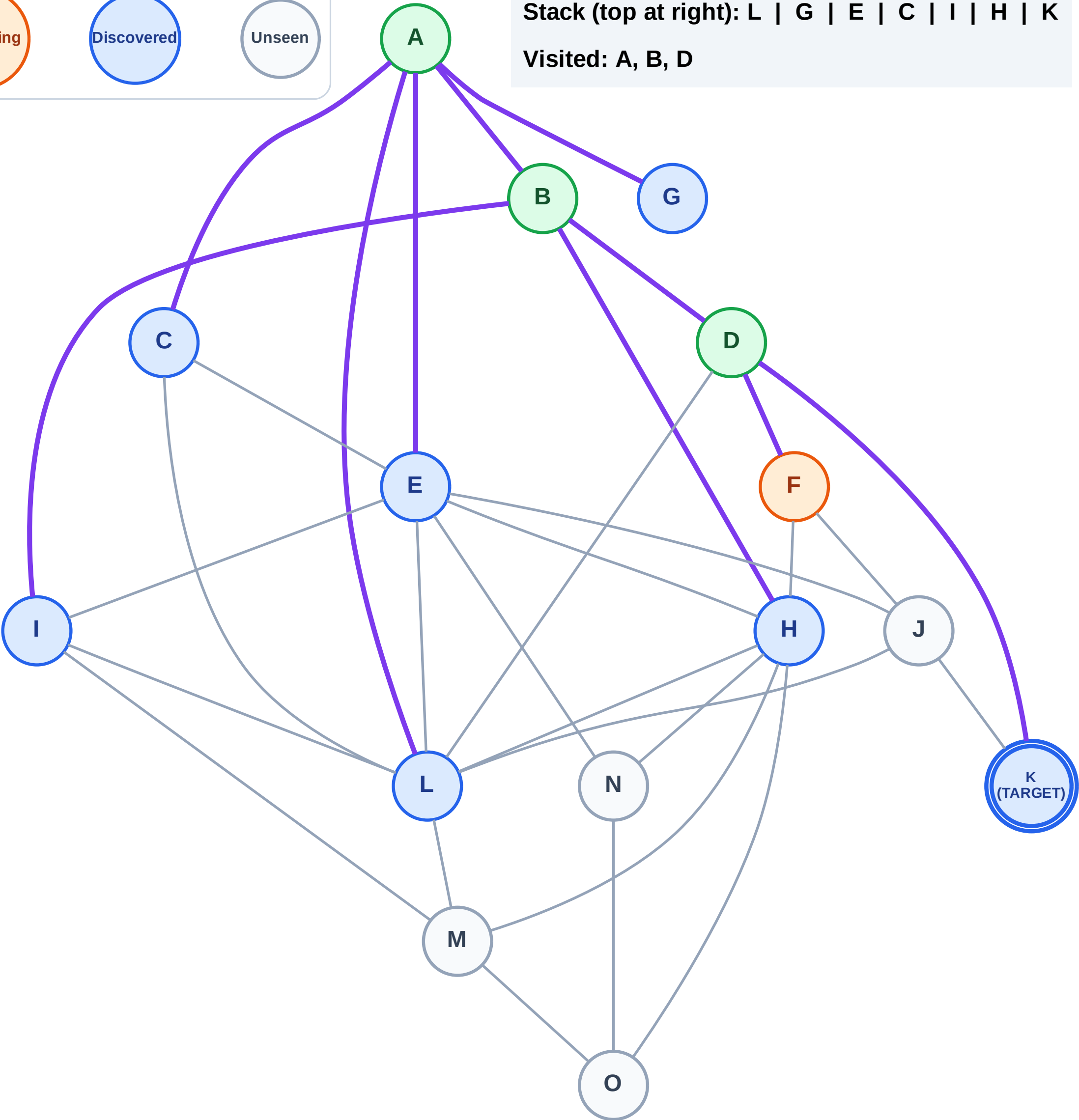
Processing

Discovered

Unseen

Stack (top at right): L | G | E | C | I | H | K

Visited: A, B, D



DFS Traversal

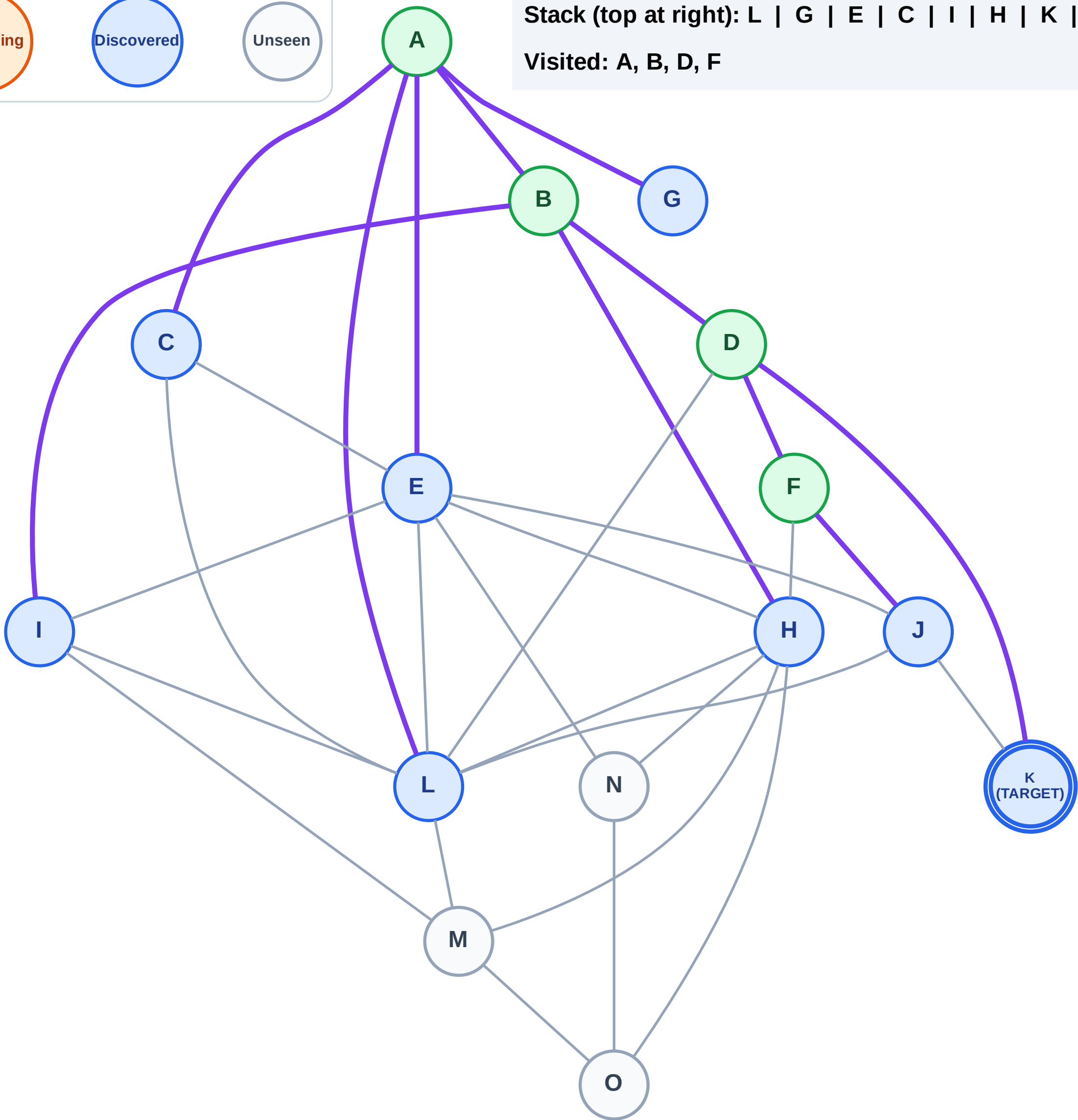
Step 9: Discover J from F

Legend



Stack (top at right): L | G | E | C | I | H | K | J

Visited: A, B, D, F



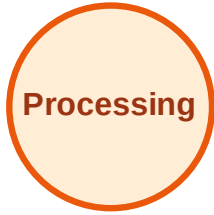
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



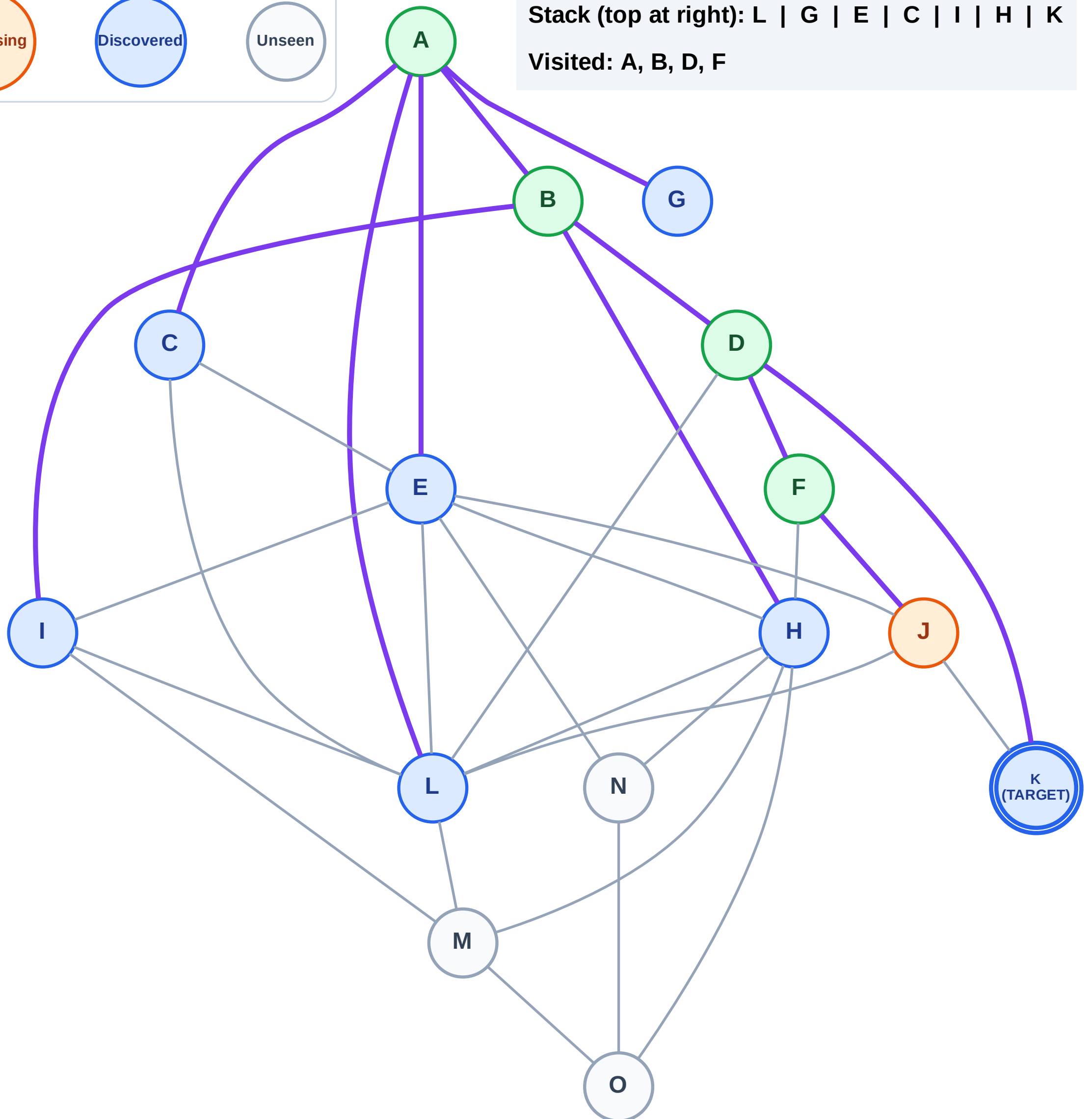
Discovered



Unseen

Stack (top at right): L | G | E | C | I | H | K

Visited: A, B, D, F



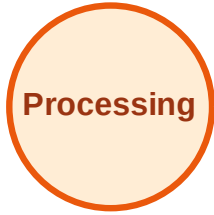
DFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



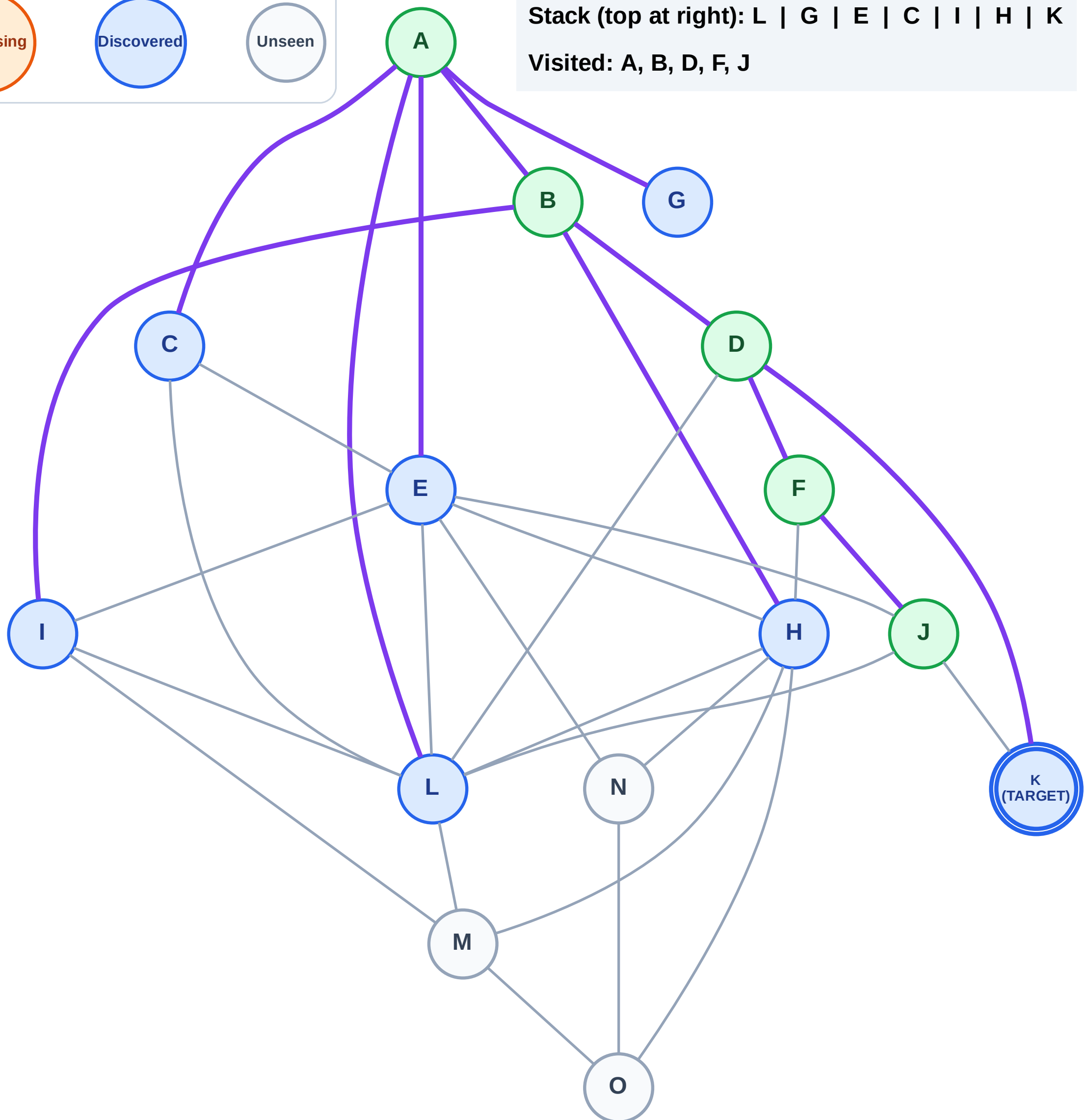
Discovered



Unseen

Stack (top at right): L | G | E | C | I | H | K

Visited: A, B, D, F, J



DFS Traversal

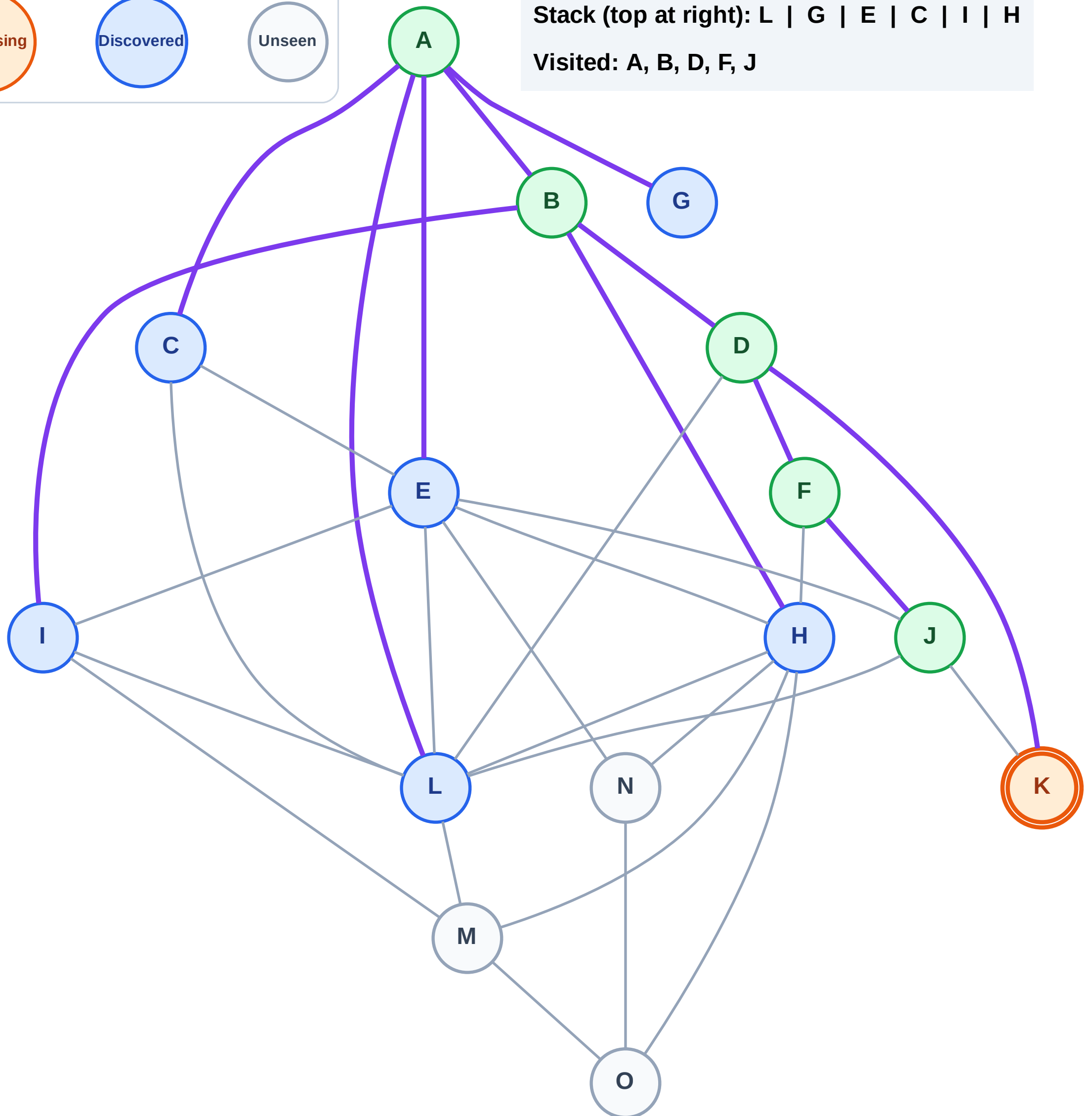
Step 12: Process node K

Legend



Stack (top at right): L | G | E | C | I | H

Visited: A, B, D, F, J



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | G | E | C | I | H

Visited: A, B, D, F, J, K

